HUMAN SPEECH BASED EMOTION RECOGNITION USING CNN & LSTM NETWORKS WITH DWT AND MFCC

A PROJECT REPORT

Submitted by

KIRTIHA V RA2011003020413

THARUNRAJ S RA2011003020414

S KEERTHANA RA2011003020458

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Dr. S. RUBIN BOSE

(Assistant Professor, Department of Computer Science and Engineering)

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RAMAPURAM, CHENNAI -600089

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SIGNATURE

Dr. S. RUBIN BOSE, M.E., M.B.A., Ph.D.,
Assistant Professor

Department of Computer Science and Engineering
SRM Institute of Science and Technology,
Ramapuram, Chennai-89

SIGNATURE

Dr. K. RAJA, M.E., Ph.D.,
Professor and Head

Department of Computer Science and Engineering
SRM Institute of Science and Technology,
Ramapuram, Chennai-89

Submitted for the Viva Voce Examination held on _____ at SRM Institute of Science and Technology, Ramapuram, Chennai -600089

INTERNAL EXAMINER **EXTERNAL EXAMINER**

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
RAMAPURAM, CHENNAI - 89

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We hereby declare that the entire work contained in this project report titled **"HUMAN SPEECH BASED EMOTION RECOGNITION USING CNN & LSTM NETWORKS WITH DWT AND MFCC"** is the bonafide work of **KIRTIHA V [RA2011003020413], THARUNRAJ. S [RA2011003020414], S. KEERTHANA [RA2011003020458]** at SRM Institute of Science and Technology, Ramapuram, Chennai - 600089, under the guidance of **Dr. S. RUBIN BOSE, Assistant Professor,** Departmentof Computer Science and Engineering.

Place: Chennai

Date:

KIRTIHA V

THARUNRAJ S

S KEERTHANA

SRM Institute of Science & Technology
Department of Computer Science and Engineering

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Student Name : Kirtiha V, Tharunraj S, S Keerthana

Registration Number : RA2011003020413, RA2011003020414, RA2011003020458

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KIRTIHA V

THARUNRAJ S

S KEERTHANA

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LIST OF ABBREVATIONS

HSER	Human Speech Emotion Recognition
CREMA	Crowd-sourced Emotional Multimodal Actors Dataset
SAVEE	Speech and Audio in the Vocal Expression
RAVDESS	Ryerson Audio-Visual Database of Emotional Speech and Song
TESS	Toronto Emotional Speech Set
DWT	Discrete Wavelet Transform
MFCC	Mel-Frequency Cepstral Coefficients
LSTM	Long Short-Term Memory
LWCNN-LSTM	Light Weight Convolutional Neural Network with Long Short-Term Memory
Resnet18-LSTM	Residual Networks18 with Long Short-Term Memory
ReLU	Rectified Linear Unit

ABSTRACT

Human Speech emotion recognition (HSER) is an important part of human-computer interaction, affective computing, and mental health assessments. In this research, we give a complete investigation of SER that uses deep learning models and wavelet transformations for feature extraction. Our research focuses on four popular emotion datasets: TESS, CREMA-D, SAVEE, and RAVDESS. We investigate the efficiency of Long Short-Term Memory (LSTM), Light Weight Convolutional Neural Network with Long Short-Term Memory (LWCNN-LSTM) and Resnet18-LSTM architectures for feature representation when combined with discrete wavelet transform (DWT) and MFCC (Mel-Frequency Cepstral Coefficients). For MFCC, we explore coefficients of dimensions 16, 24, 32, and 40, while for DWT, we study levels 1, 2, 3, 4, and 5 of the db1 family wavelet. We study the effects of different wavelet levels and MFCC coefficients on model performance. Experimental results show that the Resnet18-LSTM model consistently outperforms the LWCNN_LSTM and LSTM model on all datasets, showing outstanding accuracy and generalization capabilities. CNN-LSTM and LSTM both reached up to 99.64% and 96.96% accuracy on the TESS dataset and Resnet18-LSTM reached a remarkable 99.82% accuracy. We find that model performance improves with increased wavelet levels; Resnet18-LSTM performs better than LWCNN_LSTM and LSTM. Our results demonstrate the effectiveness of deep learning models in conjunction with wavelet transforms for human speech recognition (HSER), highlighting their potential for practical applications that need precise emotion recognition from audio data.

CHAPTER 1

INTRODUCTION

1.1 INTRODUCTION

The use of deep learning techniques has contributed significantly to the impressive advancements made in Human Speech Emotion Recognition (HSER or SER) in recent years. Popular methods for extracting important aspects from voice data includes Convolutional Neural Networks (CNNs) and Recurrent Neural Networks (RNNs), which employ particular designs such Long Short-Term Memory (LSTM) and Gated Recurrent Units (GRU). CNNs are particularly skilled at capturing the spatial elements included in spectrogram representations, even if RNNs are superior at modelling temporal dependencies in sequential data. CNNs are therefore an excellent option for evaluating voice signals with changing emotional content. HSER performance is enhanced by the synergistic coupling of Convolutional Recurrent Neural Networks (CRNNs), a hybrid design that combines the best aspects of both CNNs and RNNs. Improvements in data augmentation methods, such as voice synthesis and alteration, have been crucial in increasing the variety and depth of emotional speech datasets. Through the artificial enlargement of the emotional expression and scenario repertoire, these methods enable SER models to be trained more thoroughly, leading to enhanced generalization capabilities and increased robustness in noisy contexts.

Despite these notable advancements, there remain a number of challenges in the field of HSER. Managing noisy environments, dealing with unbalanced information, and ensuring the model's interpretability remain significant challenges that demand ongoing study and innovative solutions. These difficulties must be addressed in order to achieve the potential of emotion-aware computing systems in a range of human-machine interaction scenarios, such as virtual assistants, affective computing, and mental health monitoring applications.

This paper aims to examine and assess Discrete Wavelet Transform (DWT) and Mel-Frequency Cepstral Coefficients (MFCC) performance on human speech emotion recognition (HSER) tasks. These models are tested on a range of emotional speech datasets, such as TESS, RAVDESS, SAVEE, and CREMA-D (or CREMA).

1.2 PROBLEM STATEMENT:

Developing a robust system for emotion recognition in human speech using Resnet18-LSTM, Light Weight Convolutional Neural Networks (LWCNN) and Long Short-Term Memory (LSTM) networks. This system aims to accurately classify the emotions conveyed in audio recordings, enabling applications in fields such as affective computing, mental health monitoring, and human-computer interaction.

In recent years, there has been a growing interest in developing systems capable of accurately recognizing human emotions from speech signals. Despite advancements in machine learning and neural network architectures, achieving robust emotion recognition remains challenging due to the complex and dynamic nature of human emotions. This project aims to address this challenge by proposing a novel approach that combines the power of CNN and LSTM networks. The proposed system will be designed to process raw audio data, extract relevant features, and learn the temporal dynamics of emotional expressions using CNN and LSTM layers. By training the model on a large dataset of labeled audio recordings encompassing various emotional states, the system will be able to accurately classify emotions such as happiness, sadness, anger, fear, and neutral states. The ultimate goal of this project is to develop a reliable speech emotion recognition system that can be deployed in real-world applications.

1.3 SCOPE AND MOTIVATION

The scope of this project is to develop a system for recognizing basic human emotions from speech using CNN and LSTM networks. We will focus on recognizing a limited set of well-defined emotions, such as anger, sadness, happiness, and neutrality. We will utilize publicly available speech datasets for training and testing our models. The project will involve developing and evaluating various CNN and LSTM based architectures for emotion recognition performance, focus on recognizing basic emotions (anger, sadness, happiness, disgust, surprise), utilize publicly available datasets for training and testing, develop and evaluate CNN and LSTM based models for emotion recognition.

The motivation for this project stems from the growing interest in developing human-computer interfaces . Emotion recognition technology has the potential to significantly improve the user experience in various applications, such as customer service chatbots, virtual assistants, and educational software. Additionally, this project is motivated by the advancements in deep

learning, particularly the capabilities of LSTM networks in processing sequential data like speech signals.

- Growing interest in human-computer interaction with emotional intelligence.
- Potential applications in various domains (e.g., customer service, healthcare).
- Advancements in deep learning, particularly LSTM networks.

1.4 OBJECTIVE

The main objective is to develop a Human Speech Emotion Recognition (HSER) that achieves high accuracy in recognizing basic human emotions. Additionally, we aim to develop a robust model that can generalize well to unseen data, meaning it can perform well on data it has not been trained on before. Finally, we will explore techniques to make the model more interpretable, allowing us to understand how it makes its predictions and gain insights into the relationship between speech features and emotions. Achieve high accuracy in recognizing basic emotions from speech, develop a robust model that generalizes well to unseen data, explore the interpretability of the CNN and LSTM networks for emotion recognition.

CHAPTER 2

LITERATURE REVIEW

2.1 LITERATURE REVIEW

Loan Trinh Van et al. [1] investigated how well deep neural network architectures, specifically the Gated Recurrent Unit (GRU), for speech emotion recognition (SER) interact with the IEMOCAP corpus. The GRU model is distinguished by its persistent dominance, with respect to assessment metrics, over CNNs and CRNNs. With an accuracy of 97.47%, the GRU model that it could accurately distinguish and categorize emotional states expressed in voice signals. GRU's enhanced performance was mostly due to its ability to store knowledge over time, which also contributed to mitigating the vanishing gradient problem that deep learning systems occasionally encounter. Additionally, they investigated data augmentation methods specifically, voice alteration which produced appreciable gains in recognition efficiency. These models demonstrated greater generalization ability by increasing the variety of emotional speech data, which helped them to raise accuracy rates for all emotional classes. These findings illustrate the importance of data augmentation techniques in boosting model robustness efficiency of deep learning methods in SER.

Wani et al. [3] highlights the importance of Speech Emotion Recognition (SER) in Human-Computer Interaction (HCI) and speech processing systems, emphasizing the need for effective methodologies to extract and classify emotional features from speech signals. The author emphasizes major challenges due to the gap between human and machine perception of emotional speech cues. The paper delves into the methodologies and design features of SER systems, with a particular emphasis on the use of speech databases for training and approaches to prosodic and spectral acoustic feature extraction. The author discusses both conventional and deep learning classifiers and highlights the importance of deep learning methods to the evolution of SER.

Abbaschian et al. [5] aims to provide a comprehensive understanding of the subject by comparing deep learning and classic machine learning techniques for SER. The paper focuses on accessible datasets, methods, problems, and future directions. Their comparison shows a shift away from conventional machine learning and signal processing techniques and toward more advanced deep learning architectures like VAE (Variational Autoencoder), GAN (Generative Adversarial Network), and LSTM (Long Short-Term Memory). This study offers

the framework for future research that aims to tackle these difficulties and develop the SER discipline.

Wang et al. [6] presents a novel dual-level model for Speech Emotion Recognition (SER) called MLSTM+MDS-LSTM which integrates mel-spectrograms obtained from raw audio signals and MFCC characteristics. The author applied Preprocessing to every speech, producing two mel-spectrograms with distinct time-frequency resolutions and MFCC features. A unique Dual-Sequence LSTM (DSLSTM) architecture simultaneously processes the two mel-spectrograms while a traditional LSTM processes the MFCC characteristics. With a weighted accuracy of 72.7% and an unweighted accuracy of 73.3%, the author demonstrates how well mel-spectrograms and MFCC features work together in SER tasks. The results indicate how successful the dual-level architecture is by attaining more accuracy with fewer parameters.

Xie et al. [7] presents a new method for speech emotion recognition (SER) that uses attention-based long short-term memory (LSTM) recurrent neural networks in conjunction with frame-level speech data. Conventional techniques frequently fail to include the temporal correlation present in speech waveforms, resulting in less-than-ideal recognition precision. The suggested technique preserves temporal relations across a series of frames, makes use of frame-level speech qualities that are taken from waveforms to address this. Furthermore, the LSTM architecture incorporates attention processes to emphasize variations in emotional saturation between frames.

Experiments on various emotion corpora, such as CASIA, eNTERFACE, and GEMEP, show that the suggested method outperforms the most recent algorithms in terms of performance. When attention-weighting techniques are added to the LSTM output, the model's capacity to extract emotional data pertinent to classification tasks is accelerated, which improves stability on the test set and speeds up training convergence. Compared to typical LSTM models, the LSTM-TF model. Notable improvements are evident when analysing the quantitative recognition accuracy of various emotions in each test set, especially when it comes to differentiating subtle emotions. Overall performance increases have been seen in CASIA and eNTERFACE corpora for the LSTM-TF models, with improved performance across all emotions except rage.

Kurpukdee et al. [8] present a novel technique for Speech Emotion Recognition that use a convolutional long short-term memory recurrent neural network (ConvLSTM-RNN) to extract phoneme-based features from raw input speech signals. The proposed model improves

recognition accuracy by extracting phoneme-based emotion probabilities from each frame of an input utterance and converting them into statistical features for emotion classification at the utterance level using support vector machines (SVMs) or linear discriminant analysis (LDA) systems. Four essential emotions were tested experimentally using the Interactive Emotional Dyadic Motion Capture (IEMOCAP) database: neutral, happy, sad, and wrath. The proposed method outperforms the baseline system by using 192 phoneme-based emotion probabilities to reach an accuracy of 57.46%, outperforming standard ConvLSTM-based systems. The author effectively extracted phoneme-based features by utilizing ConvLSTM-RNN, which enhanced recognition accuracy for a variety of emotions. The results revealed that the proposed approaches greatly increased performance; in both the 192 phoneme-based and 36 phoneme-class-based emotions, SVMs and LDA classifiers outperformed the baseline system by 4.45% to 8.28%. Happy is found to be the most challenging emotion to classify. Anger was the most easily recognized emotion, followed by neutral and grief, with SVM employing the polynomial kernel achieving the highest accuracy (65.13%).

Rasheed et al. [9] proposed a System for Automatic Speech Emotion Recognition (ASER) to recognize and extract relevant information from voice signals in order to classify different emotional categories. The author included a relation-aware attention system into an attention-based framework that is offered to get over these restrictions. This framework focuses on both short-term and long-term temporal data combines the advantages of CNNs and Bi-LSTM networks. In order to improve performance accuracy even more, the author suggested a method with CNN+BiLSTM frameworks that rely on attention. The suggested method beat previous approaches while keeping processing speeds comparable, according to the data. To be more precise, the suggested strategy outperformed earlier approaches with an average accuracy of 90.25% while little affecting processing times.

Susithra et al. [10] studied the efficacy and computing efficiency of Mel Frequency Cepstral Coefficients (MFCC) for speech signal processing are highlighted in this paper's introduction and implementation using Simulink. The author collected MFCC characteristics and used a recurrent neural network (RNN) for word detection in speech signals. The MFCC feature extraction is applied to further speech signal processing by its smooth integration with RNN. The paper emphasizes the importance of feature extraction in speech recognition and the efficacy of MFCC in capturing spectral characteristics.

Niu et al. [11] investigates the use of RNN sequence algorithms in digital signal processing on computers and to address challenging signal processing problems. RNNs are shown to be more effective than traditional approaches, particularly in voice recognition and real-time video processing. The author states that the combination of AI and machine learning technologies has the potential to significantly increase digital signal processing, opening the door to novel applications in fields like autonomous cars and medical diagnostics, despite obstacles like computational needs.

Onah et al. [12] investigates the effectiveness of CNN and LSTM networks for Speech Emotion Recognition (SER). Utilizing the RAVDESS dataset, LSTM models are configured with three layers (70, 50, 60 units), followed by dense layers (48 units with ReLU activation, 14 units) and CNN models comprise Conv1D layers with increasing filters (1024, 512, 256, 128) and max-pooling, followed by a GlobalAverage-Pooling1D layer and dense layers (64 units, ReLU, 0.2 dropout, 14 units with softmax). Training involved 50 epochs for LSTM and 100 epochs for CNN, both with Adam optimizer and mean squared error loss calculation. Hyperparameters like batch size, learning rate, and epochs were tuned to optimize model performance, with a learning rate of 0.001 yielding improved accuracy. Experimental results demonstrate CNN's training accuracy of 97% and test accuracy of 64%, while CNN+LSTM convergence swiftly to optimal performance, showcasing a significant improvement in SER tasks.

Leelavathi et al. [13] studied structured audio data in conjunction with LSTM networks to detect and classify emotions in speech signals. The author suggested LSTM-based optimized approaches as combining techniques to reduce processing expenses and make use of attention mechanisms to retrieve task-related information. To avoid overfitting, the model is evaluated using graphs that show accuracy and loss against epochs. With 78.2% accuracy in predicting the right emotions, simulation results using the RAVDESS dataset meet the project's goal of recognizing emotions with LSTM-based recurrent neural networks.

Table 2.1 Summary of the existing works

Author & Year	Parameter	Dataset	Model	Accuracy/ Precision (%)
Trinh et al **[1]**	Spectrogram, Mel-frequency	IEMOCAP	CNN CRNN GRU	96.69 97.18 97.47
Wang Et al **[6]**	Mel-Spectrograms MFCC	IEMOCAP	Dual-Sequence LSTM and CNN	73.3
Xie et al **[7]**	Mel-frequency	CASIA eNTERFACE GEMEP	Attention Based LSTM	92.82 89.97 56.04
kurpukdee et al **[8]**	Frame level ad utterance level	IEMOCAP	Conv- LSTM	Anger - 68.72 Happiness – 61.04 Neutral – 67.04 Sadness- 64.67
Rasheed et al **[9]**	min-quiet interval, and min-sounding interval	RAVDESS IEMOCAP	CNN Bi-LSTM	83.16 87.41
Susithra et al **[10]**	MFCC	-	LSTM	-
Niu et al **[11]**	DSP Sequence classification	-	RNN	-
Onah et al **[12]**	Mel-frequency MFCC	RAVDESS	LSTM and CNN	97
Leelavathi et al **[13]**	MFCC	RAVDESS	LSTM	78.2
Xiaoyu et al **[15]**	MFCC	IEMOCAP	CNN- BiLSTM	90.65

Table 2.1 summarizes the findings of several studies on the accuracy of speech emotion recognition algorithms. The studies employed a variety of datasets, including IEMOCAP , RAVDESS, CASIA, and ENTERFACE. The features used to represent speech also varied across studies, and included spectrograms, Mel-frequency , MFCCs and gammatone filterbank

features. Finally, the speech emotion recognition algorithms implemented in the studies included CNNs, RNNs, LSTM networks, and combinations of these architectures.

2.2 PROPOSED WORK

In this research endeavor, we embark on a comprehensive exploration of speech emotion recognition employing advanced signal processing techniques. Our methodology is anchored in the acquisition of a robust speech dataset, meticulously preprocessed using the DWT and MFCC to distill pertinent features. The ensuing dataset undergoes a meticulous partitioning into distinct subsets for rigorous testing and training, facilitating thorough evaluation. Central to our investigation is the evaluation of three distinct neural network architectures: the Resnet18-LSTM, the LWCNN-LSTM hybrid and the LSTM model. Leveraging the extracted features from both MFCC and DWT, these architectures are primed to discern emotions embedded within speech signals. We meticulously scrutinize the efficacy of each architecture in capturing temporal nuances and extracting discriminative features through rigorous training and evaluation processes. This study aims to push the boundaries of speech emotion recognition by amalgamating cutting-edge signal processing methodologies with state-of-the-art neural network architectures, fostering advancements in human-computer interaction and emotional intelligence modeling.

This section details the proposed system for developing a speech emotion recognition system

1. **Data Collection and Preparation**
    - A diverse speech dataset encompassing various emotions and speakers.
    - Each speech sample needs clear emotional labels assigned by human annotators.
2. **Preprocessing and Feature Extraction**
    - Discrete Wavelet Transform (DWT) is used for preprocessing.
    - Preprocessing ensures consistency across recordings
    - The preprocessed dataset is further split into training and validation test sets.
    - Relevant features are extracted to capture emotional information within the speech signal using Mel-Frequency Cepstral Coefficients (MFCCs) .
3. **Model Architecture Design**
    - A CNN and LSTM based neural network architecture is designed. CNNs excel at extracting local features, while LSTMs handle the sequential nature of speech data.
    - Experimentation with different CNN-LSTM configurations optimizes performance.

4. **Model Training and Optimization**

- The model is trained on the training set using a suitable optimizer and loss function.
- Techniques like learning rate scheduling and regularization are explored for optimization.

5. **Model Evaluation**

- The trained model is evaluated on the unseen test set for unbiased assessment.
- Metrics include accuracy, precision, recall, and F1-score are calculated for each emotional category.
- The model's performance is compared with baseline models to assess the deep learning approach's effectiveness

2.3 SYSTEM SPECIFICATIONS

2.3.1 Software Requirements

1. Python
2. PyTorch
3. Keras
4. Scikit-learn
5. Librosa
6. Jupyter Notebook

2.3.2 Hardware Requirements

1. CPU
2. GPU (optional but recommended)
3. RAM (>8gb)
4. Storage(for storing)
5. Microphone (for audio input)

CHAPTER 3

MODULE DESCRIPTION

3.1 METHODOLOGY

The methodology of HSER (Figure 3.1) involves acquiring a speech dataset and preprocessing it using the DWT and MFCC to extract relevant features. The dataset is then divided into subgroups for testing and training in order to be evaluated. In this work, we investigate the performance of three different neural network architectures: Resnet18-LSTM, LWCNN-LSTM and LSTM. The retrieved features from MFCC and DWT are used to train these architectures, which enable them to recognize emotions from speech signals. We evaluate how well each architecture captures temporal correlations and extracts discriminative features for emotion recognition through rigorous training and evaluation.

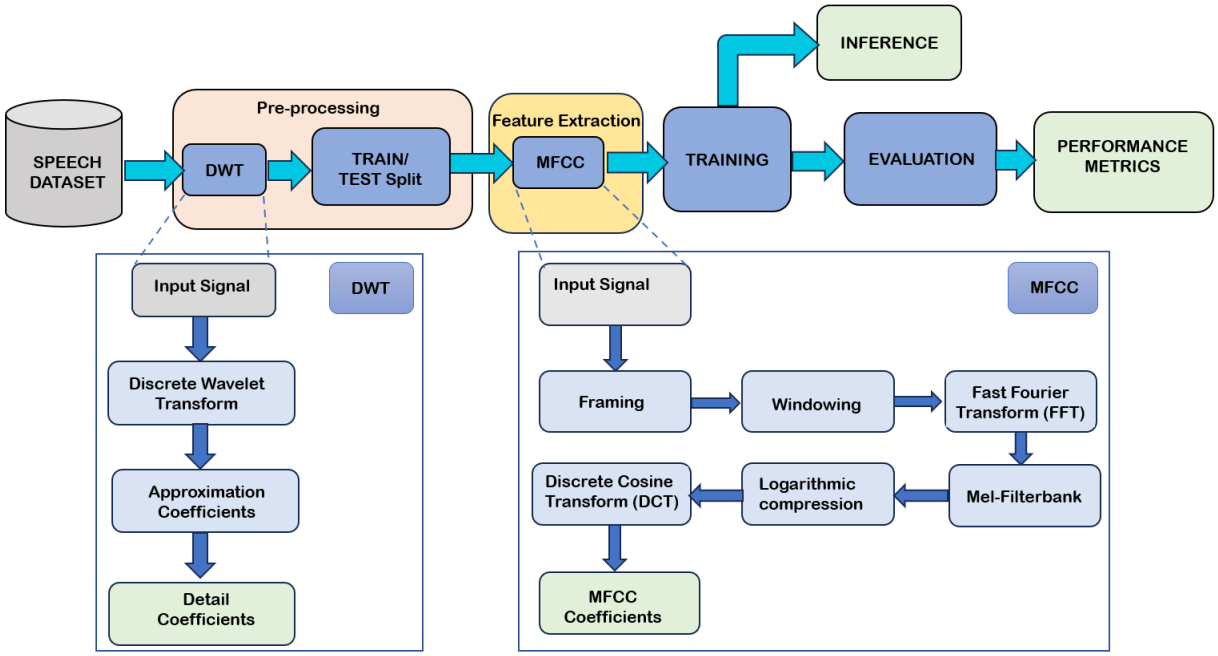


Fig 3.1 Functional Block Diagram of HSER

3.1.1 Discrete Wavelet Transform (DWT)

The "Discrete Wavelet Transform"(DWT)[2] is a powerful signal analysis tool that has gained considerable traction in various fields, including speech processing and emotion recognition. Unlike its predecessor, the Short Time Fourier Transform (STFT), DWT offers a distinct advantage in its ability to provide both high time resolution and frequency localization, making it to captures transient features within signals. This characteristic mirrors the behaviour

of the human auditory system, which demonstrates varying sensitivity to frequency components over time, akin to the time-frequency resolution properties exhibited by the DWT.

At the heart of the DWT lies a pyramidal algorithm that decomposes a signal into its constituent frequency bands, each containing information about different levels of detail. By successively applying highpass and lowpass filtering operations, the DWT extracts both approximation and detail coefficients, allowing for a hierarchical representation of the signal across different resolutions. This decomposition process, coupled with the downsampling operation, ensures that the number of resulting wavelet coefficients remains consistent with the input signal size, facilitating efficient analysis and feature extraction.

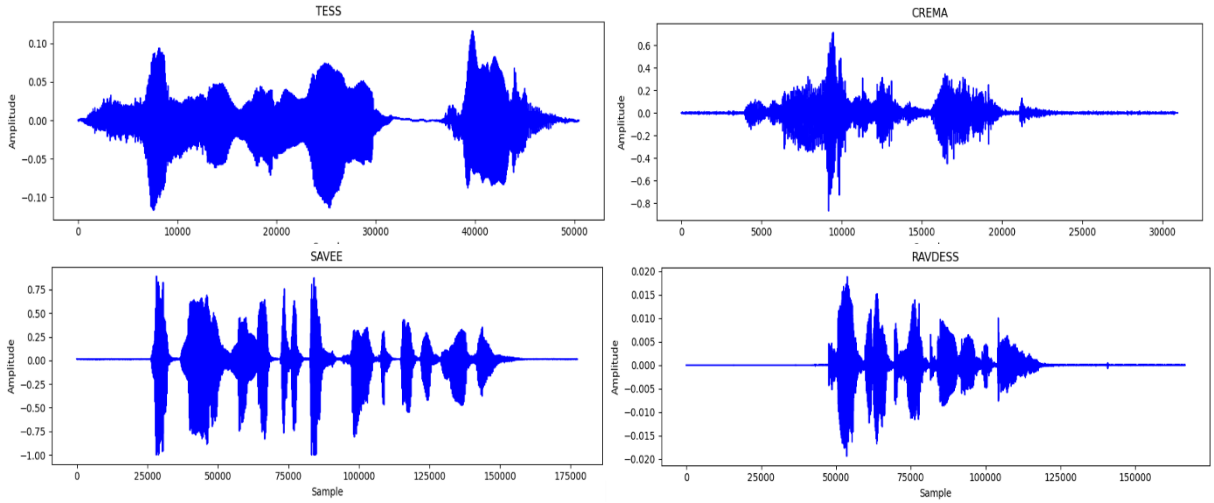


Fig 3.1.1 Graphical representation of original signals

Figure 3.1.1 depicts the raw speech signal as a waveform. The x-axis represents time, and the y-axis represents amplitude of the signal. This raw waveform contains information corresponding to the spoken words, but it's not directly interpretable by a machine learning model. The subsequent stages of the speech recognition system will extract features from this waveform that are suitable for training the deep learning model

Let $x[n]$ be the input signal, and $h[n]$ and $g[n]$ be the low-pass and high-pass analysis filters, respectively.

$$Approximation\ coefficients\ A_j[n] at\ scale\ j:$$

$$A_j[n]=\sum_k h[k] \cdot A_{j+1}[2n - k] \quad \text{.....(1)}$$

$$Detail\ coefficients\ D_j[n] at\ scale\ j:$$

$$D_j[n]=\sum_k g[k] \cdot A_{j+1}[2n - k] \quad \text{.....(2)}$$

Where:

- $A_j[n]\ and\ D_j[n]$ are the approximation and detail coefficients at scale j.
- $A_{j+1}[n]$ is the approximation coefficients at the next finer scale.
- $h[n]\ and\ g[n]$ are the analysis filters.

3.1.2 Mel-Frequency Cepstral Coefficients (MFCC)

The "Mel-Frequency Cepstral Coefficients" (MFCC) is another core in speech signal processing, providing a concise yet informative description of speech features. MFCC computing, which was inspired by the nonlinear frequency perception of the human auditory system, consists of a succession of well-defined steps aimed at capturing critical spectrum information while minimizing the impact of irrelevant noise and distortions.

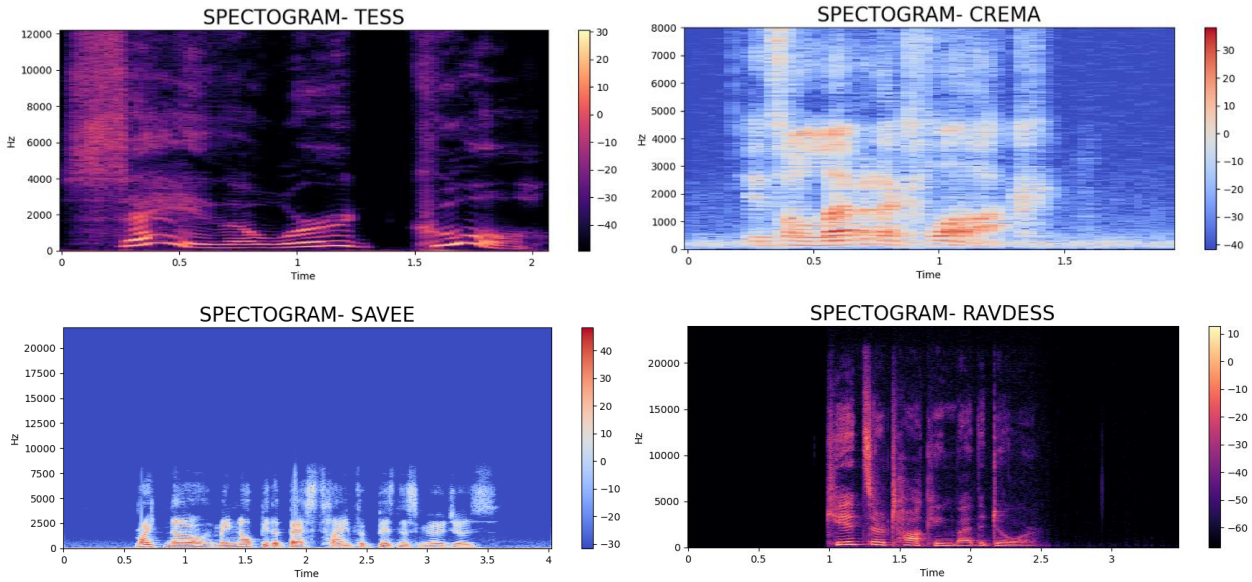


Fig 3.1.2 Spectrograms of original signals

Figure 3.1.2 display spectrograms of speech signals from different datasets. Spectrograms are visual representations of the frequency content of a signal over time. Darker areas indicate higher energy at a specific frequency and time. The process begins with framing, where the speech signal is divided into short, overlapping segments to facilitate analysis in the frequency domain.

$$The\ Speech\ Signal\ y[n]\ is\ divided\ into\ frames\ of\ fixed\ duration$$

Subsequently, each frame undergoes windowing to minimize spectral leakage and ensure coherence during Fourier analysis.

$$Discrete\ Fourier\ Transform\ (DFT)$$

$$X[k] = DFT(y[n] \cdot w[n]) \quad \text{.....(3)}$$

- $X[k]$ is the Discrete Fourier Transform of the windows signal
- $y[n]$ is the pre-emphasized signal
- $w[n]$ is the window function

The resulting magnitude spectrum is then transformed using a Mel-filterbank, which effectively mimics the nonlinear frequency resolution of the human ear. By emphasizing critical regions of the spectrum while attenuating less perceptually relevant components, the Mel-filterbank operation aligns MFCC computation with human auditory perception.

$$Mel - filterbank:$$

$$H_m[k] = \sum_{i=1}^{N} H_m^i[k] \quad \text{.....(4)}$$

- $H_m[k]$ represents the output of the Mel-filterbank
- $H_m^i[k]$ represents the response of the i-th triangular filter in the Mel-scale

Following Mel-filterbanking, the logarithm of the filterbank outputs is computed to approximate the human auditory system's nonlinear response to sound intensity.

$$M_m[k] = \log(H_m[k]) \quad \text{.....(5)}$$

- $M_m[k]$ is the logarithm of the filterbank outputs
- $H_m[k]$ is the output of the Mel-filterbank

This logarithmic compression enhances the discriminative power of MFCC features by emphasizing differences in spectral energy across frequency bands. Finally, the Discrete Cosine Transform (DCT) is applied to the logarithmically scaled filterbank outputs, yielding a set of MFCC coefficients that encapsulate the essential spectral characteristics of the speech signal.

$$Discrete\ Cosine\ Transform(DCT):$$

$$c_i = \sum_{k=1}^{K} M_m[k] \cdot \cos[\frac{\pi i}{K}\left(k - \frac{1}{2}\right)] \quad \text{.....(6)}$$

- $M_m[k]$ is the logarithm of the filterbank outputs
- c_i represents the MFCC coefficients

We use MFCC as a core feature extraction strategy, for its ability to capture speech-specific information while lowering dimensionality. By introducing MFCC-based features into our emotion detection system, we hope to capitalize on the discriminative strength of spectral characteristics in detecting minor fluctuations in emotional expression. MFCC's resistance to external noise and channel unpredictability makes it an essential tool for real-world applications in speech signal processing and emotion identification research.

3.2 MODEL ARCHITECTURE

3.2.1 LSTM

The LSTM network a type of Recurrent Neural Network (RNN) architecture designed to effectively capture long-range dependencies in sequential data, making it particularly suitable for tasks such as speech emotion recognition.

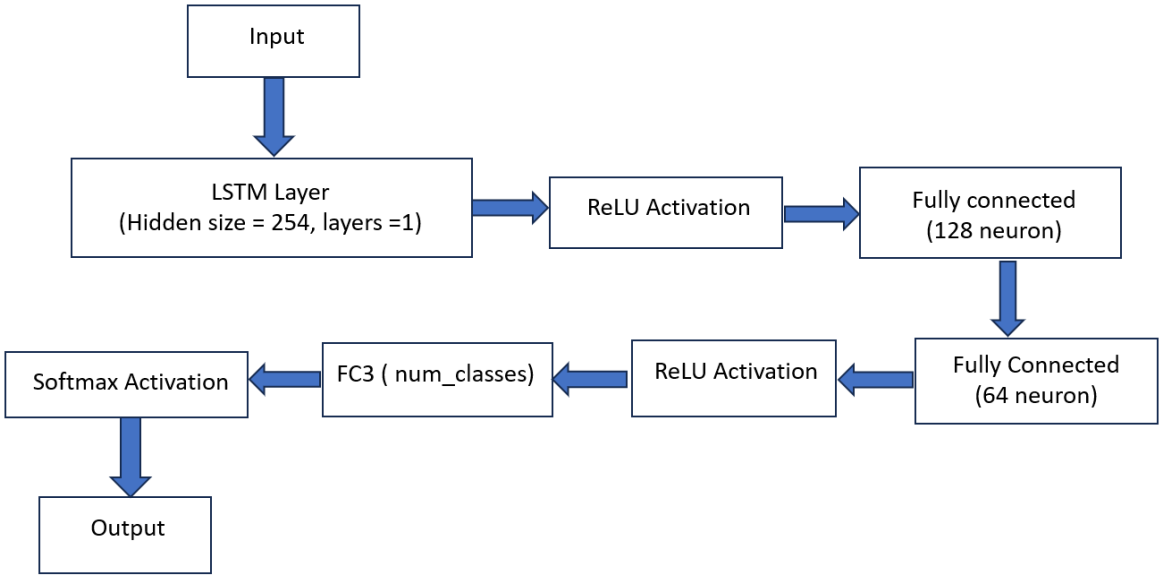


Fig 3.2.1 Architectural Block Diagram of LSTM model

The architecture of the LSTM model(Figure 3.2.1) for speech emotion identification consists of numerous interconnected layers designed to efficiently process and interpret sequential speech input. At the centre of the model is the LSTM layer made up of stacked LSTM cells. These cells are specialized recurrent neural network units capable of preserving information over lengthy sequences critical for detecting temporal relationships in speech signals. Each LSTM cell stores hidden states, allowing the model to learn and remember patterns over numerous time steps. Following the LSTM layer, fully connected (FC) layers are

used to extract additional high-level features and representations from the sequential data. “ReLU” activation functions are implemented after each FC layer to add nonlinearity and help the model learn complex mappings between input characteristics and emotional states. The final output layer, which includes a softmax activation function, generates the model's predictions by modelling the probability distribution across distinct emotion classes. This architecture enables the model to effectively learn and recognize emotional cues in voice signals, providing useful insights into speakers' underlying emotional states.

3.2.2 LWCNN-LSTM

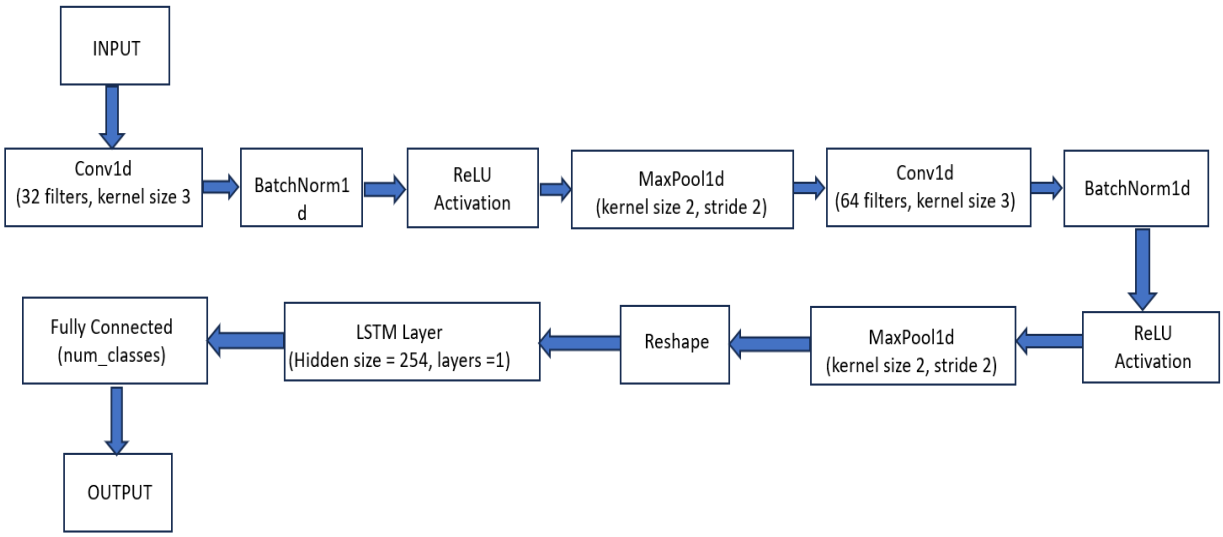


Fig 3.2.2 Architectural Block Diagram of LWCNN-LSTM model

The LWCNN-LSTM model architecture (Figure 3.2.2) for speech emotion recognition is designed to efficiently analyse sequential speech input capturing both local and temporal connections. The LWCNN-LSTM model combines CNN and LSTM networks designed to interpret sequential input such as audio. Initially, two 1D convolutional layers are used, followed by batch normalization to improve training stability. These convolutional layers are critical in obtaining hierarchical information from raw audio sources. Each convolutional operation is followed by ReLU (or “Rectified Linear Unit”) activation functions, which allow for the insertion of non-linearity to capture subtle patterns within the input. Max-pooling layers then downsample the feature maps while retaining important features, resulting in dimensionality reduction.

Following the convolutional layers, the model proceeds to an LSTM layer, which is capable of capturing temporal dependencies and long-range patterns in sequential data.

Configured with a specific number of hidden units and layers, the LSTM layer efficiently learns from the sequential nature of the incoming audio data. The output of the LSTM layer is then fed into a fully connected layer, which results in a softmax activation function. This final layer provides probability distributions for all output classes, ensuring that the model's predictions are normalized probabilities.

This architecture effectively combines the skills of CNN in feature extraction from audio sources with the sequence modelling capabilities of LSTM networks. While CNN layers extract hierarchical characteristics, the LSTM layer processes them across time, capturing temporal dependencies and making predictions. The model can capture complicated patterns in the data by including “ReLU” activation functions, and the “Softmax “activation guarantees that the final output is acceptable for classification tasks, providing probabilistic interpretations for each class label.

3.2.3 Resnet18-LSTM

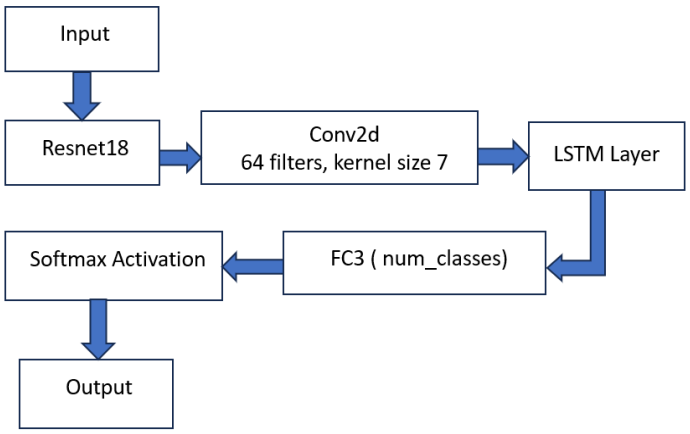


Fig 3.2.3 Architectural Block Diagram of Resnet18-LSTM model

The integration of a ResNet-18 backbone with an LSTM sequence model (Figure 3.2.3) represents a novel approach to speech emotion recognition, offering a robust architecture capable of extracting hierarchical features from audio data while effectively capturing temporal dependencies. Leveraging the pre-trained ResNet-18 model as a feature extractor provides a solid foundation for encoding high-level representations from raw audio input. By adapting the ResNet-18 architecture and redefining its initial convolutional layer to accommodate mono-channel audio, the model can effectively process sequential audio data.

Following the feature extraction stage, the LSTM sequence model efficiently captures temporal dynamics inherent in speech signals. Configured with a single layer and hidden size

of 128, the LSTM layer adeptly learns from the sequential nature of the extracted features, thereby enabling the model to discern nuanced patterns and long-range dependencies within the input sequences. This fusion of convolutional and recurrent architectures allows for a comprehensive analysis of both local and global information, enhancing the model's ability to discern emotional cues embedded within speech.

The final classification head, comprising a linear layer, produces probability distributions over the target emotion classes. By leveraging the output of the LSTM layer, the model generates predictions that are informed by both the hierarchical representations learned by the ResNet-18 backbone and the temporal dynamics captured by the LSTM. The utilization of a “Softmax” activation ensures that the model's predictions are normalized probabilities, facilitating interpretable outputs suitable for classification tasks.

Overall, this ResNet-18-LSTM architecture represents a sophisticated solution for speech emotion recognition, effectively combining the strengths of convolutional feature extraction with the sequential modeling capabilities of LSTM networks. Through this integrated approach, the model demonstrates a capacity to discern complex emotional patterns in speech data, thereby holding promise for applications in affective computing and human-computer interaction domains.

CHAPTER 4

EXPERIMENTAL SETUP

4.1 EXPERIMENTAL SETUP

4.1.1 DATA COLLECTION

The dataset collection process for the emotion recognition task involved gathering audio recordings from four distinct datasets: CREMA, RAVDESS, SAVEE, and TESS. Each dataset offers a unique contribution to the diversity and richness of emotions captured in speech signals, enabling comprehensive model training and evaluation.

Table 4.1 Description of HSER Datasets

Emotion	**TESS**	**CREMA**	**SAVEE**	**RAVDESS**
ANGER	400	1271	60	192
CALM	-	-	-	192
DISGUST	400	1271	60	192
FEAR	400	1271	60	192
HAPPY	400	1271	60	192
NEUTRAL	400	1087	120	96
SAD	400	1271	120	192
SURPRISED	400	-	-	192
TOTAL	***2800***	***7442***	***480***	***1440***

Table 4.1 provides a detailed breakdown of the emotion distribution across the four speech emotion recognition datasets used in this study (TESS, CREMA, RAVDESS and SAVEE)

The TESS (or “Toronto Emotional Speech Set,” dataset)[16] contains recordings of North American English speakers expressing a variety of emotions in a controlled environment. TESS includes 400 samples in each of seven emotional categories: neutral, happy, sad, anger, fear, disgust, surprised. The addition of multiple speakers and emotional expressions increases the dataset's richness and relevance in emotion recognition investigations.

The CREMA or “Crowd Emotion Multimodal Acted,” dataset [17] contains audio recordings of professional actors expressing various emotions. With over 7,000 annotated clips, CREMA provides a diverse range of emotional emotions: happy, sad, anger, fear, disgust, neutral. These recordings offer a controlled and regulated setting for researching emotional cues in speech.

SAVEE (or “Surrey Audio-Visual Expressed Emotion”) dataset[18], contains audio recordings of male speakers expressing brief words with various emotional expressions. Happy, anger, sad, neutral, disgust, and fear are represented in SAVEE, providing a compact yet valuable resource for emotion recognition research.

RAVDESS (or “Ryerson Audio-Visual Database of Emotional Speech and Song,”) dataset[19] has 192 speech recordings from actors executing scripted scenarios expressing eight major emotions: happy, calm, sad, angry, fear, disgust, neutral and surprised. With a total of 24 actors contributing to the dataset, RAVDESS captures nuanced emotional states across a diverse set of voices and performances.

4.1.2 DATA PREPROCESSING AND FEATURE EXTRACTION

Preprocessing and feature extraction are critical steps in preparing raw audio data for further analysis and model training. This step provides a comprehensive preprocessing pipeline for four distinct emotion recognition datasets: CREMA, RAVDESS, SAVEE, and TESS. Each dataset undergoes similar preprocessing steps, including data extraction from file names, organization into standard emotion categories, and storage in CSV files. Throughout the process the consistency is maintained by mapping emotion labels to standard categories and showing file distribution among emotion. Then the files are organized into folders based on their emotion labels, facilitating easy access and further analysis. By systematically preparing the datasets and standardizing emotion labels, the preprocessing pipeline establishes a solid foundation for subsequent tasks such as feature extraction and model training in emotion recognition applications.

Then feature extraction is started by loading the dataset CSV files for each of the four datasets: RAVDESS, SAVEE, CREMA, and TESS. These CSV files contain the file paths and corresponding emotion labels for the audio recordings. Pandas python library is used to read these files into Data Frame objects for easy manipulation. The distribution of emotion labels in each dataset is visualised to gain insights into the dataset's composition. This stage is counting

the occurrences of each emotion label, which allows us to assess the dataset's balance and diversity in emotional expressions. To prepare the emotion labels for model training, one-hot encoding is used to convert categorical emotion labels into binary matrix representations. This encoding approach makes categorization easier by portraying each emotion as a binary vector, with each element corresponding to a distinct emotion label. Then the feature extraction using the Discrete Wavelet Transform (DWT) and Mel-Frequency Cepstral Coefficients (MFCC) techniques is performed. The MFCC features capture relevant information from the audio signals, providing rich representations of speech data across different wavelet levels.

The Dataset after DWT is split into train and test sets and with mfcc coefficients as the extracted features (X) and their corresponding emotion labels (y) is saved into separate NumPy arrays. These preprocessed data arrays are organized into directories named after each dataset, ensuring systematic storage and easy access during model training and evaluation. The preprocessing and feature extraction pipeline lays the groundwork for building robust emotion recognition models capable of effectively analysing speech data from diverse sources.

4.1.3 TRAINING

The training process is done in a Laptop with AMD Ryzen7 processor with 16gb ram and integrated radeon graphics. The Training begins by loading preprocessed data, which includes input features extracted using the Discrete Wavelet Transform (DWT) with a level of 1,2,3,4,5 and Mel-frequency cepstral coefficients (MFCC) with feature dimensions of 16, 24, 32, 40, along with their corresponding labels. The dataset is split into training and validation sets using a specified test size, ensuring an adequate distribution of data for model training and evaluation. PyTorch Data Loader is then employed to create iterable data loaders for batch-wise processing during training and validation, optimizing computational efficiency.

4.1.3.1 LSTM MODEL

The LSTM model architecture for speech emotion recognition comprises an input size of 1, a hidden size of 256, and a single LSTM layer. Following the LSTM layer, there are three fully connected layers with “ReLU” activation functions. The first fully connected layer takes the output of the LSTM layer, which has a hidden size of 256, and reduces it to 128 neurons. The output is further reduced to 64 neurons in the second fully connected layer before reaching the final layer, which outputs probabilities for each emotion class using a softmax activation

function. This architecture is designed to capture temporal dependencies in the input features and learn hierarchical representations for effective emotion recognition.

4.1.3.2 LWCNN-LSTM MODEL

The LWCNN-LSTM model architecture is designed with flexibility and performance in mind. It begins with a one-dimensional convolutional neural network (CNN) comprising two convolutional layers followed by “Batch Normalization” and “ReLU” activation functions to extract hierarchical features from the input spectrogram. The first convolutional layer consists of 32 filters with a kernel size of 3, while the second layer comprises 64 filters with the same kernel size 3. “Max-pooling” layers with a kernel size of 2 are applied after each convolutional layer to downsample the feature maps.

The output from the CNN layers is fed into a LSTM layer with 256 hidden units and one layer, facilitating the model's ability to capture temporal dependencies in the spectrogram features. The LSTM layer processes the input sequences and generates output sequences, which are then passed through a fully connected layer with “Softmax” activation to produce class probabilities. The model utilizes the “Cross-Entropy Loss” function and the “Adam” optimizer with a learning rate of 0.001 for training, chosen based on prior experimentation to balance convergence speed and stability.

4.1.3.3 RESNET18-LSTM MODEL

The Resnet18-LSTM model architecture is meticulously crafted to harness the power of both convolutional and recurrent neural networks for effective speech emotion recognition. Beginning with a ResNet18 backbone the model adapts its convolutional layers to accommodate mono-channel audio input. The first convolutional layer, with a kernel size of (7, 7), strides through the spectrogram, capturing intricate spatial features crucial for emotion detection. Followed by Batch Normalization and “ReLU” activation, these convolutional layers transform raw audio input into hierarchical representations.

Transitioning seamlessly into sequential analysis, the architecture seamlessly integrates an LSTM layer, configured with 128 hidden units, designed to capture temporal dependencies within the spectrogram features. As the LSTM processes input sequences, it unravels the dynamic temporal patterns inherent in speech signals, enriching the model's understanding of emotional nuances over time. This fusion of convolutional and recurrent architectures ensures that the model can effectively discern subtle emotional cues embedded within speech.

Culminating in the classification head, the LSTM output is distilled into actionable insights through a fully connected layer with softmax activation. This final layer generates class probabilities, providing interpretable outputs for classification tasks. By leveraging the ResNet18 backbone and LSTM sequence modeling, the architecture achieves a delicate balance between feature extraction and temporal analysis.

Algorithm 1: LSTM Model Training for Emotion Recognition from Speech Signals

Input: $NC_i =$ Number of MFCC coefficients, $D[n] =$ Level of Discrete Wavelet Transform
Output: Trained LSTM model, Training and validation performance metrics.

```
for each input:
    def preprocess(input, D[n])
        Load the data.
        D_[x] = Discrete Wavelet Transform (DWT) with (D[n]) to decompose the speech signals into coefficients
        actual class(y) = class of D_[x]
        return D_[x], y
    def extract_features(D_[x],NC_i):)
        Extract NC_i [Mel-frequency cepstral coefficients (MFCC)]
        Calculate the mean of MFCC coefficients to obtain feature vectors representing the speech signals
        return X = array(mfcc_coeff)
    if LSTM
        model = LSTM()
    elseif LWCNN-LSTM()
        model = LWCNN-LSTM()
    elseif Resnet18-LSTM()
        model = Resnet18-LSTM()
    def train_model(X ,y):
        criterion(Loss) = CrossEntropyLoss
        optimizer(Opt) = Adam(lr=0.001)
        metrics = ['accuracy', 'Precision', 'Recall','F1-Score']
        train_data(D_[t]) = 0.80X samples
        validation_data(D_[v]) = 0.20X samples
        for each in epochs:
            model.train= (Training data =D_[t] , Optimizer =Opt, LOSS = Loss)
      def validate(validation_data):
          for inputs, labels in validation_data:
              output_predictions (O_p) = predicted_class
    def inference, predictions=[ ] :
        for predictedn in output_prediction:
            max_index = prediction.argmax()
        if max_index == 0:
            inference_results.append('anger')
        elseif max_index == 1:
            inference_results.append('calm')
        elseif max_index == 2:
            inference_results.append('disgust')
        elseif max_index == 3:
            inference_results.append('fear')
        elseif max_index == 4:
            inference_results.append('happy')
        elseif max_index == 5:
            inference_results.append('neutral')
        elseif max_index == 6:
            inference_results.append('sad')
        elseif max_index == 7:
            inference_results.append('surprised')
```

4.1.4 EVALUATION

To assess the performance of the HSER models several metrics including “precision”, “recall”, “accuracy” and “f1-score” are employed. These metrics provide insights into the models' ability to correctly classify emotional states from audio data.

Precision (“also known as positive predictive value”) measures the proportion of correctly predicted positive instances among all instances predicted as positive by the model.

$$Precision = \frac{True\ Positives}{True\ Positives + False\ Positives} \quad \text{.....(7)}$$

Recall (“also known as sensitivity or true positive rate”), measures the proportion of actual positive instances that are correctly identified by the model.

$$Recall = \frac{True\ Positives}{True\ Positives + False\ Negatives} \quad \text{.....(8)}$$

Accuracy (“also known as the overall correctness of the model's predictions) is calculated as the ratio of correctly classified instances to the total number of instances.

$$Accuracy = \frac{True\ Positives + True\ Negatives}{True\ Positives + True\ Negatives + False\ Positives + False\ Negatives} \quad \text{.....(9)}$$

The F1-score is a harmonic mean of precision and recall, providing a single metric that balances both precision and recall.

$$F1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \quad \text{.....(10)}$$

For HSER models, these metrics are calculated based on the model's predictions compared to ground truth labels across the emotional categories such as anger, calm, disgust, fear, happy, neutral, surprised, and sad. “True positives” (TP) are instances where the model correctly predicts the presence of a specific emotion, “False positives” (FP) are instances where the model incorrectly predicts the presence of an emotion, “False negatives” (FN) represent instances where the model fails to detect the presence of an emotion, and “True negatives” (TN) are instances correctly classified as not belonging to a particular emotional category. By computing precision, recall, accuracy and accuracy for the HSER models across different emotional categories, we gain a comprehensive understanding of their performance in identifying human emotional states from speech audio data.

CHAPTER 5

RESULTS AND DISCUSSIONS

5.1 ANALYSIS ON FEATURE EXTRACTION TECHNIQUES

5.1.1 DWT LEVEL

The impact of different wavelet levels (DWT 1-5) on speech emotion recognition is explored by experimenting with N_MFCC set to 40 on all four datasets.

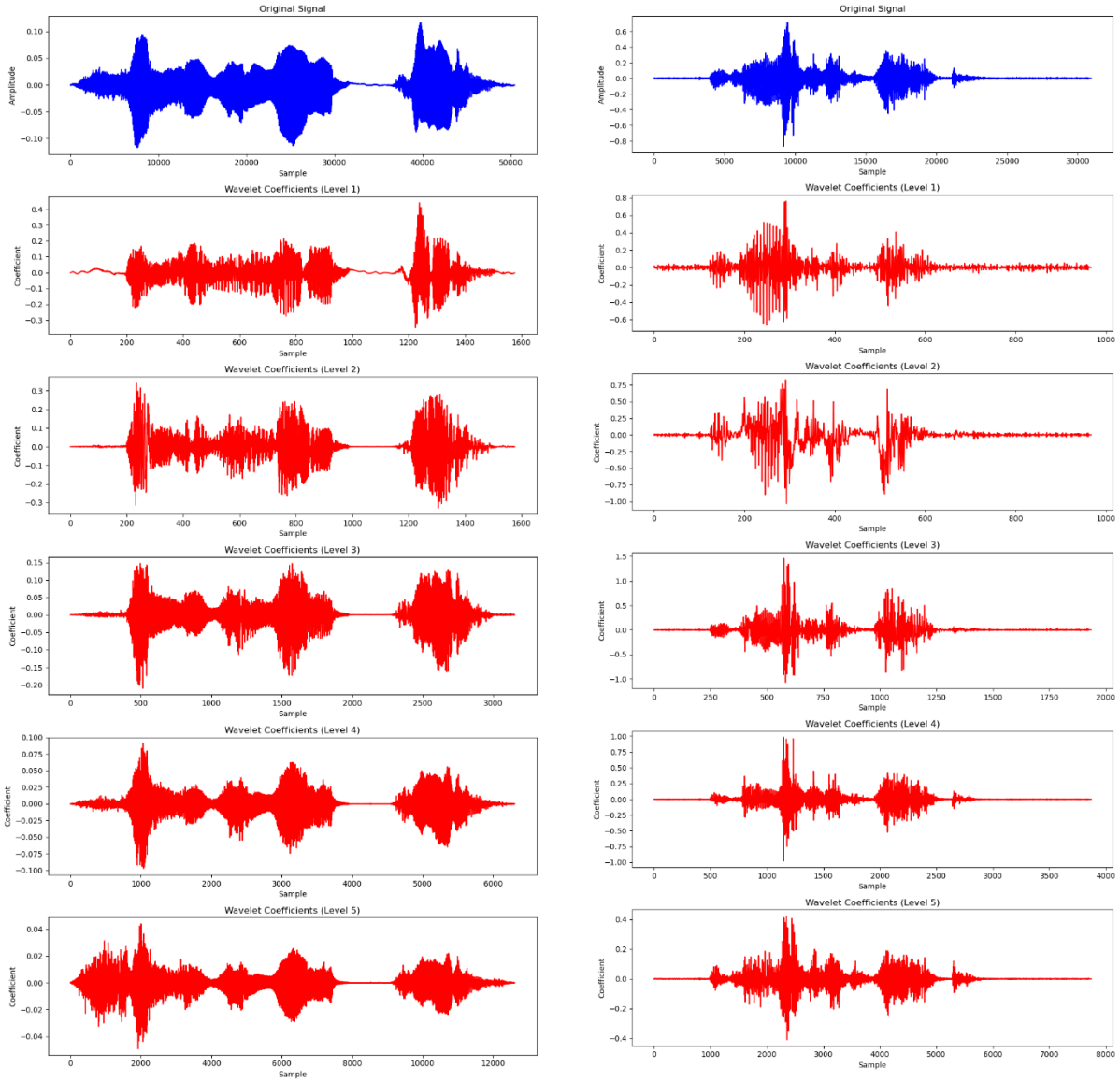


Fig 5.1.1 Graphical representation of signals after DWT

Figure 5.1.2 depict the wavelet coefficients of the speech signal after undergoing Discrete Wavelet Transform (DWT). DWT decomposes the signal into various sub bands of

frequency and time resolution using wavelet functions. The original signal top signal in (Figure 5.1.1) is decomposed into a single approximation coefficient set (cA1) and a set of detail coefficient sets (cD1,cD2, ...). The low-frequency components of the signal are captured by the approximation coefficients , while the detail coefficients capture the high-frequency components at different scales. Fig. 5.1.1 shows the wavelet coefficients after decomposition one by one after each level. Here, the cA1 represents the approximation coefficients and the cD1 represents the detail coefficients. As we can see from the figure, most of the energy is concentrated in the approximation coefficients, indicating the dominant low-frequency content in the speech signal. Fig. 5.1.1 illustrates that with more decomposition levels, the detail information about the high-frequency components of the signal is captured in more detail across multiple detail coefficient sets (cD1, cD2, ..., cD5).

A. MODEL TRAINING

Table 5.1.1 Performance metrics with 40 MFCC coefficients

Dataset	DWT	LSTM				LWCNN-LSTM				RESNET18-LSTM			
		Acc (%)	Pre (%)	Rec (%)	F1S (%)	Acc (%)	Pre (%)	Rec (%)	F1S (%)	Acc (%)	Pre (%)	Rec (%)	F1S (%)
TESS	1	99.2	99.2	99.2	99.2	99.96	99.96	99.96	99.96	99.91	99.91	99.91	99.91
	2	99.6	99.6	99.6	99.6	99.87	99.87	99.87	99.87	99.96	99.96	99.96	99.96
	3	**99.42**	**99.42**	**99.42**	**99.42**	**99.96**	**99.96**	**99.96**	**99.96**	**100**	**100**	**100**	**100**
	4	99.11	99.11	99.11	99.11	99.91	99.91	99.91	99.91	**100**	**100**	**100**	**100**
	5	**99.42**	**99.42**	**99.42**	**99.42**	**99.96**	**99.96**	**99.96**	**99.96**	99.96	99.96	99.96	99.96
CREMA	1	81.62	81.83	81.62	81.61	85.37	85.37	85.37	85.33	85.37	85.37	85.37	85.33
	2	64.29	64.27	64.29	63.84	87.42	87.51	87.42	87.39	86.72	87.61	86.72	86.53
	3	**81.64**	**81.87**	**81.64**	**81.57**	85.45	85.47	85.45	85.47	**87.5**	**89.09**	**87.5**	**87.4**
	4	82.87	82.95	82.87	82.8	**89.08**	**89.14**	**89.08**	**89.07**	98.18	98.18	98.18	98.18
	5	82.18	82.5	82.18	82.18	87.1	87.22	87.1	87.05	87.1	87.22	87.1	87.05
SAVEE	**1**	**81.77**	**83.73**	**81.77**	**81.52**	86.72	87.61	86.72	86.53	**100**	**100**	**100**	**100**
	2	78.65	80.35	78.65	78.13	87.5	89.09	87.5	87.4	**100**	**100**	**100**	**100**
	3	76.04	75.97	76.04	75.27	87.76	88.44	87.76	87.68	97.76	98.44	97.76	97.68
	4	70.57	63.44	70.57	65.68	84.38	85.36	84.38	83.73	84.38	85.36	84.38	83.73
	5	75	76.02	75	74.23	**90.36**	**91.35**	**90.36**	**90.35**	90.36	91.35	90.36	90.35
RAVDESS	1	80.12	80.41	80.12	80	82.47	82.83	82.47	82.43	88.37	88.77	88.37	88.28
	2	79.25	79.47	79.25	79.04	**88.37**	**88.77**	**88.37**	**88.28**	**100**	**100**	**100**	**100**
	3	**83.42**	**83.69**	**83.42**	**83.32**	78.21	73.92	78.21	75.56	99.42	99.42	99.42	99.42
	4	72.48	73.14	72.48	71.77	87.85	88.21	87.85	87.76	87.85	88.21	87.85	87.76
	5	77.69	78.41	77.69	77.41	67.01	58.17	67.01	61.44	87.76	88.44	87.76	87.68

***Acc- Accuracy, *Pre- Precision, *Rec-Recall, *F1S- F1-Score**

Table 5.1.1 presents the performance of HSER models on different DWT levels. The The wavelet level selection affects the model's capacity to extract emotional characteristics

from speech input. Higher DWT levels generally produce superior results in terms of recall, precision, F1-score and accuracy; this is particularly evident in the TESS dataset. As seen by the CREMA dataset, where performance peaks at DWT level 4 for LSTM but somewhat decreases for LWCNN-LSTM, this tendency may not hold true for all datasets. Likewise, in the RAVDESS dataset, DWT level 3 has the maximum recall, precision, and LSTM accuracy, although its F1-score is lower than that of DWT level 2.

These differences highlight how crucial it is to choose the number of MFCC coefficients and wavelet level carefully in order to achieve the best results possible on speech emotion identification tasks. We find that, in order to maximize model performance in speech emotion recognition tasks, N_MFCC should be kept at 40 on the TESS dataset.

With training accuracy and other metrics continuously above 99%, the model obtains remarkable performance metrics for the TESS dataset (N_MFCC value 40). This shows that throughout a range of feature dimensions and training times, the model trained on TESS data can successfully represent the subtleties of emotional speech. On the CREMA, SAVEE, and RAVDESS datasets, on the other hand, the model's performance varies greatly. There are significant differences between the model's performance on TESS and these datasets, despite the model showing moderate to good performance overall, with training accuracy ranging from roughly 60% to 80%. This suggests that changes in data quality, recording settings, and emotional expressions may provide problems for the model when it comes to generalizing across datasets.

B. MODEL EVALUATION

Based on the training it can be concluded that employing Discrete Wavelet Transform (DWT) at level 3 provides a balance between retaining important features and minimizing noise. Further decomposition beyond level 3, as shown in cD4 and cD5, may introduce excessive noise and miss finer features present in the original signal. Conversely, employing only level 1 of DWT may fail to sufficiently capture the intricate details of the signal. Hence, DWT level 3 emerges as an optimal choice, striking a balance between feature retention and noise suppression in the wavelet domain representation of the speech signal.

Table 5.1.2 Performance metrics with 40 MFCC coefficients

Dataset	DWT	LSTM				LWCNN-LSTM				RESNET18-LSTM			
		Acc (%)	Pre (%)	Rec (%)	F1S (%)	Acc (%)	Pre (%)	Rec (%)	F1S (%)	Acc (%)	Pre (%)	Rec (%)	F1S (%)
TESS	1	96.96	96.99	96.96	96.96	98.93	98.93	98.93	98.93	99.64	99.64	99.64	99.64
	2	96.96	97.02	96.96	96.96	99.29	99.29	99.29	99.28	99.46	99.46	99.46	99.46
	3	97.14	97.14	97.14	97.13	**99.64**	**99.64**	**99.64**	**99.64**	**99.82**	**99.82**	**99.82**	**99.82**
	4	**97.68**	**97.68**	**97.68**	**97.68**	99.11	99.13	99.11	99.11	**99.82**	**99.82**	**99.82**	**99.82**
	5	96.43	96.42	96.43	96.42	99.29	99.29	99.29	99.29	99.64	99.64	99.64	99.64
CREMA	1	43.72	43.02	43.72	43.1	47.28	47.01	47.28	46.52	48.09	47.72	48.09	47.59
	2	41.97	40.03	41.97	40.47	**48.96**	**48.51**	**48.96**	**47.8**	**51.37**	**50.76**	**51.37**	**50.59**
	3	**44.93**	**43.96**	**44.93**	**43.98**	47.95	47.25	47.95	47.35	51.04	50.48	51.04	50.48
	4	43.79	42.68	43.79	42.84	48.09	47.72	48.09	47.59	50.69	49.11	50.69	48.87
	5	42.65	41.82	42.65	41.65	45.94	44.95	45.94	44.68	47.28	47.01	47.28	46.52
SAVEE	**1**	59.38	55.12	59.38	55.69	**71.88**	**70.95**	**71.88**	**71.11**	79.16	79.43	79.16	78.71
	2	**60.42**	**54.3**	**60.42**	**55.81**	70.83	71.53	70.83	69.49	**80.20**	**82.83**	**80.20**	**80.47**
	3	**60.42**	**55.9**	**60.42**	**57.33**	70.83	69.78	70.83	69.89	75.69	78.21	75.69	75.61
	4	57.29	49.08	57.29	52.63	67.71	60.1	67.71	62.39	70.83	71.53	70.83	69.49
	5	**60.42**	**62.6**	**60.42**	**59.19**	69.79	69.16	69.79	67.58	70.83	71.53	70.83	69.49
RAVDESS	**1**	39.24	36.47	39.24	37.31	**59.03**	**60.16**	**59.03**	**58.78**	**68.40**	**68.80**	**68.40**	**68.23**
	2	36.11	34.84	36.11	34.17	49.65	50.44	49.65	48.82	64.12	64.68	64.12	64.01
	3	**42.71**	**40.25**	**42.71**	**40.5**	50.69	49.11	50.69	48.87	63.19	64.73	64.12	64.01
	4	39.24	37.47	39.24	36.87	51.39	52.01	51.39	50.61	59.03	60.16	59.03	58.78
	5	40.28	38.6	40.28	38.65	42.36	33.11	42.36	36.75	51.04	50.48	51.04	50.48

Table 5.1.2 showcases the validation metrics at 500 epochs for different DWT levels, while maintaining 40 MFCC coefficients, across four distinct datasets: TESS, CREMA, SAVEE, and RAVDESS. Across all datasets, varying DWT levels exhibit differing impacts on model performance. For instance, in the TESS dataset, the highest accuracy and other metrics are achieved at DWT level 4 for all the LSTM, LWCNN-LSTM and Resnet18-LSTM models, whereas in CREMA and SAVEE datasets, DWT level 3 generally yields the best results. Notably, the performance trends vary across datasets, suggesting that the optimal DWT level may be influenced by dataset characteristics. Additionally, for some datasets like RAVDESS, the performance improvements with increasing DWT levels are not as significant, indicating potential limitations or dataset-specific challenges. These findings underscore the importance of careful consideration of DWT levels in conjunction with dataset characteristics to optimize speech emotion recognition models effectively.

Overall, these results underscore the importance of dataset selection and feature representation in the development of robust speech emotion recognition models. The superior performance of the Resnet18-LSTM model on the TESS dataset highlights its potential for real-world applications requiring accurate and reliable emotion classification from speech data.

5.1.2 MFCC COEFFICIENTS

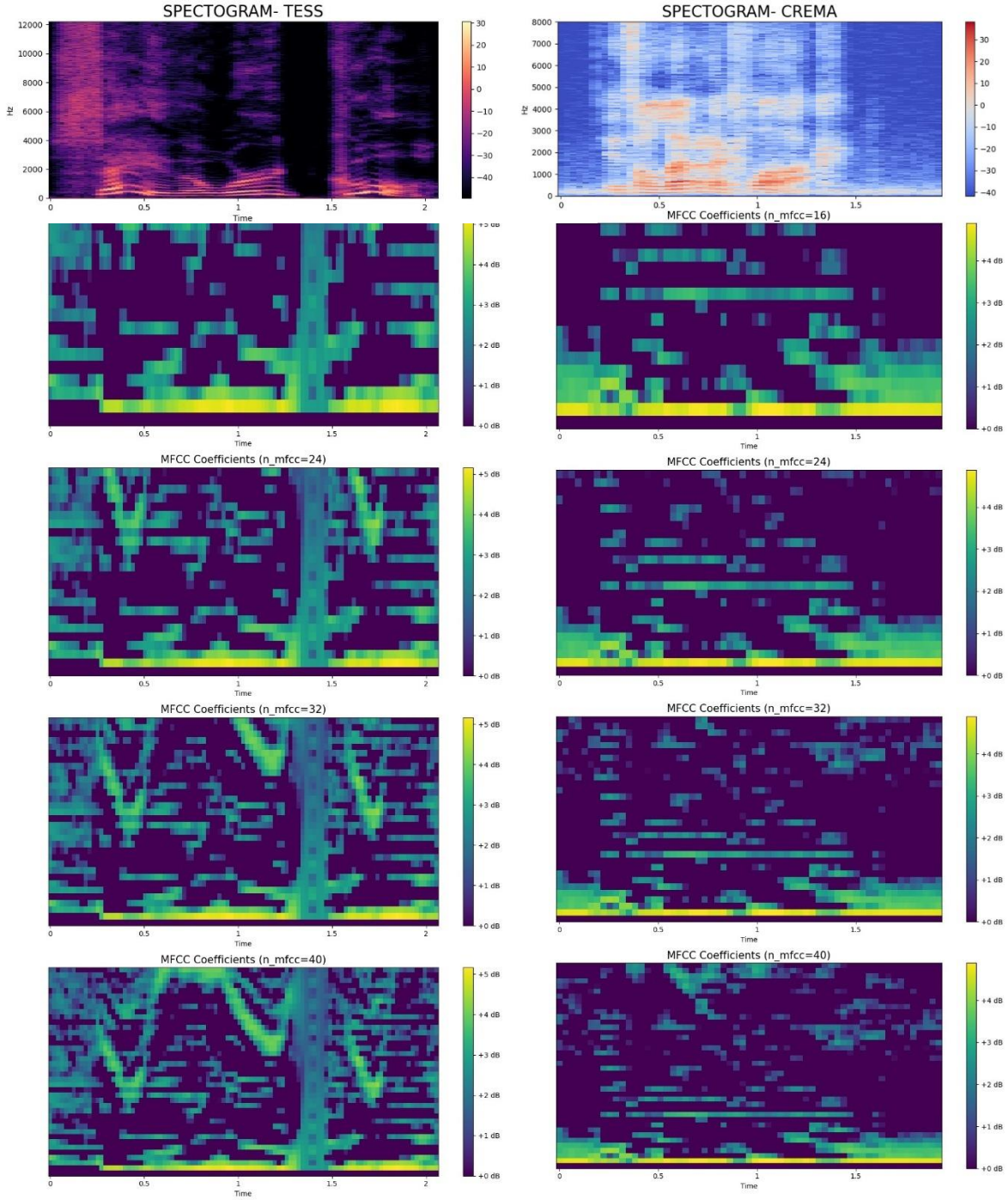


Fig 5.1.2 Spectrograms of Signals after MFCC

Figure 5.1.2 illustrate the Mel-frequency cepstral coefficients (MFCCs) derived from a speech signal. The x-axis represents the time frames (in milliseconds), while the y-axis represents

the MFCC values. The Top figure (n_mfcc=16) utilizes 16 filter banks, while the last figure (n_mfcc=40) employs 40 filter banks. A higher number of filter banks can capture finer details in the spectral envelope of the speech signal, potentially leading to improved recognition accuracy, particularly for complex sounds. However, using more filter banks also increases the dimensionality of the feature vector, which can lead to higher computational costs during training.

A. MODEL TRAINING

In addition to assessing the model's performance across different datasets and epochs, an exploration was conducted to evaluate the impact of varying the number of MFCC coefficients (N_MFCC) on the model's effectiveness in speech emotion recognition. A thorough examination of the model's behaviour with various feature representations was carried out by testing with N_MFCC values of 16, 24, 32, and 40 on the TESS dataset without using DWT preprocessing.

Table 5.1.3 Performance metrics of HSER models

Dataset	MFCC	LSTM				LWCNN-LSTM				RESNET18-LSTM			
		Acc (%)	Pre (%)	Rec (%)	F1S (%)	Acc (%)	Pre (%)	Rec (%)	F1S (%)	Acc (%)	Pre (%)	Rec (%)	F1S (%)
TESS	16	99.46	99.47	99.46	99.46	99.78	99.78	99.78	99.78	99.96	99.96	99.96	99.96
	24	99.87	99.87	99.87	99.87	99.96	99.96	99.96	99.96	**100**	**100**	**100**	**100**
	32	99.82	99.82	99.82	99.82	99.96	99.96	99.96	99.96	**100**	**100**	**100**	**100**
	40	**99.91**	**99.91**	**99.91**	**99.91**	**100**	**100**	**100**	**100**	**100**	**100**	**100**	**100**
CREMA	16	83.37	83.63	83.37	83.29	90.69	90.74	90.69	90.67	87.42	87.51	87.42	87.39
	24	81.3	81.46	81.3	81.17	90.98	91.01	90.98	90.96	**99.74**	**99.74**	**99.74**	**99.74**
	32	**86.93**	**87.01**	**86.93**	**86.91**	**91.5**	**91.52**	**91.5**	**91.48**	98.18	98.18	98.18	98.18
	40	64.52	64.78	64.52	64.46	90.29	90.35	90.29	90.26	87.1	87.22	87.1	87.05
SAVEE	**16**	**90.1**	**90.84**	**90.1**	**90.06**	94.53	94.53	94.53	94.5	**99.73**	**99.74**	**99.73**	**99.73**
	24	82.55	82.75	82.55	82.5	**96.35**	**96.43**	**96.35**	**96.36**	99.70	99.70	99.70	99.70
	32	80.73	81.67	80.73	79.45	87.76	89.5	87.76	85.5	**99.73**	**99.74**	**99.73**	**99.73**
	40	84.38	85.55	84.38	84.16	85.94	87.1	85.94	85.56	99.46	99.47	99.46	99.46
RAVDESS	**16**	**88.98**	**89.75**	**88.98**	**89.13**	92.62	92.72	92.62	92.52	**100**	**100**	**100**	**100**
	24	84.38	85.7	84.38	84.33	**96.79**	**96.82**	**96.79**	**96.76**	**100**	**100**	**100**	**100**
	32	78.82	79.31	78.82	78.64	92.88	93.1	92.88	92.34	99.91	99.91	99.91	99.19
	40	81.16	81.15	81.16	81.12	96.35	96.5	96.35	96.34	99.64	99.64	99.64	99.64

Table 5.1.3 presents the performance of HSER models. The goal of this study was to determine the ideal arrangement of MFCC coefficients to optimize model performance without adding extra complexity from DWT. Through a methodical analysis of the F1-score, recall, training accuracy, and precision metrics across these several feature dimensions, valuable

information was obtained regarding the impact of MFCC parameter selection on the LSTM model's capacity to extract pertinent emotional cues from voice data. This kind of analysis is essential for improving feature representations and adjusting model hyperparameters to maximize performance in voice emotion recognition tasks.

According to these results, using higher MFCC coefficients makes it possible to capture a more complete picture of the underlying emotional characteristics seen in speech data. The model trained on N_MFCC 40 demonstrated improved discriminative capability by utilizing a richer feature space, which allowed it to more accurately and robustly distinguish between various emotional states in TESS dataset

By using N_MFCC 32, the model is able to extract subtle emotional cues that may be essential for accurate emotion recognition by retaining more detailed spectral information. As a result, N_MFCC 32 continues to be a good option for maximizing the model's performance in speech emotion identification tasks since it strikes a compromise between computational efficiency and feature richness.

B. MODEL EVALUATION

The model with the best model state based on validation loss is saved, ensuring that the model with the optimal performance is retained for further evaluation. The confusion matrix is then generated to visualize the model's performance on the validation set, offering a clear understanding of the model's ability to classify different emotion categories accurately. The progressive improvement of the model's parameters aims to minimize the loss function while maximizing performance metrics such as precision and F1-score, facilitating efficient speech emotion recognition and guaranteeing robust generalization to inaccessible data.

The results of training the models on the TESS, CREMA, SAVEE, and RAVDESS datasets with different configurations of MFCC features and epochs reveal a variety of performance inclinations. Notably, the LSTM model consistently demonstrates high validation accuracy, precision, recall, and F1-score on the TESS dataset across all N_MFCC values, with particularly exceptional performance when N_MFCC is set to 24 - 32. Notably, the LSTM model consistently demonstrates high However, performance on the CREMA, SAVEE, and RAVDESS datasets varies significantly, with validation metrics generally lower compared to TESS. Further experiments with different wavelet levels on the TESS dataset highlight the

Resnet18-LSTM model's superiority, achieving perfect validation accuracies for select wavelet levels and outperforming the LSTM model in precision, recall, and F1-scores.

Table 5.1.4 Performance metrics of HSER models

Dataset	MFCC	LSTM				LWCNN-LSTM				RESNET18-LSTM			
		Acc (%)	Pre (%)	Rec (%)	F1S (%)	Acc (%)	Pre (%)	Rec (%)	F1S (%)	Acc (%)	Pre (%)	Rec (%)	F1S (%)
TESS	16	92.68	92.94	92.68	92.73	95.18	95.19	95.18	95.18	97.67	97.71	97.67	97.68
	24	97.14	97.17	97.14	97.15	99.46	99.47	99.46	99.46	99.46	99.47	99.46	99.46
	32	98.93	98.94	98.93	98.93	**99.64**	**99.65**	**99.64**	**99.64**	**99.64**	**99.65**	**99.64**	**99.64**
	40	**99.29**	**99.29**	**99.29**	**99.29**	99.46	99.48	99.46	99.46	**100**	**100**	**100**	**100**
CREMA	**16**	**46.47**	**46.02**	**46.47**	**45.6**	48.35	48.03	48.35	47.31	76.04	75.97	76.04	75.27
	24	45.27	44.65	45.27	44.84	47.35	47	47.35	46.96	79.16	79.43	79.16	78.71
	32	42.04	40.9	42.04	40.83	**48.62**	**48.27**	**48.62**	**47.66**	**80.20**	**82.83**	**80.20**	**80.47**
	40	42.38	42.16	42.38	41.22	48.15	48.2	48.15	47.51	75.69	78.21	75.69	75.61
SAVEE	**16**	60.42	60.36	60.42	59.03	**70.83**	**70.24**	**70.83**	**70.2**	**72.91**	**73.81**	**72.91**	**73.06**
	24	**64.58**	**66.32**	**64.58**	**62.71**	68.75	69.31	68.75	68.23	70.13	69.91	69.44	69.63
	32	59.38	64.23	59.38	57.13	66.67	65.79	66.67	63.91	69.79	69.16	69.79	67.58
	40	57.29	52.55	57.29	54.25	**70.83**	**73.64**	**70.83**	**69.82**	70.13	69.91	69.44	69.63
RAVDESS	**16**	**50.69**	**51.33**	**50.69**	**50.38**	62.15	61.92	62.15	61.24	69.44	70.31	69.44	69.63
	24	42.01	43.28	42.01	40.83	**67.01**	**68.1**	**67.01**	**66.81**	**71.52**	**71.92**	**71.52**	**70.96**
	32	39.24	38.4	39.24	37.69	64.93	64.71	64.93	63.93	70.13	69.91	69.44	69.63
	40	37.85	39.23	37.85	38.01	64.58	64.66	64.58	64.2	70.83	69.78	70.83	69.89

Table 5.1.4 presents the validation metrics obtained at 500 epochs for different MFCC coefficient values across four datasets: TESS, CREMA, SAVEE, and RAVDESS. For the TESS dataset, as the N_MFCC value increases from 16 to 40, there is a consistent improvement in accuracy, precision, recall, and F1-score for all the HSER models. However, for datasets like CREMA and RAVDESS, the performance improvements are not as pronounced compared to TESS and SAVEE, suggesting potential dataset-specific factors influencing model performance. These results underscore the importance of optimizing MFCC coefficient values in speech emotion recognition tasks and highlight the need for dataset-specific considerations when designing and evaluating models.

5.2 RESULTS AND DISCUSSION

5.2.1 RESULTS AND FINDINGS

Experiments were conducted using four diverse datasets: CREMA, RAVDESS, SAVEE, and TESS, each offering unique emotional expressions for comprehensive model training and evaluation. Analysis of varying N_MFCC and different wavelet levels (DWT)

revealed that higher DWT levels generally improved model performance, albeit with variations across datasets and model architectures. Validation results highlighted the LWCNN-LSTM and LSTM model's consistent high performance on the TESS dataset, while Resnet18-LSTM model outperformed both the models in precision, recall, and F1-score, particularly on TESS with varying DWT levels**.**

Table 5.2.1 Performance of LSTM metrics on each dataset

DATASET	MFCC	DWT	TRAINING				VALIDATION			
			Acc (%)	Pre (%)	Rec (%)	F1S (%)	Acc (%)	Pre (%)	Rec (%)	F1S (%)
TESS	40	2	99.59	99.59	99.59	99.59	96.96	97.01	96.96	96.96
CREMA	40	4	82.86	82.95	82.86	82.80	43.78	42.67	43.78	42.83
SAVEE	16	1	84.89	85.60	84.89	84.06	62.50	67.77	62.50	62.80
RAVDESS	40	3	83.42	83.69	83.42	83.31	42.70	40.25	42.70	40.49

In Table 5.2.1, we examine the performance of LSTM models on each dataset, focusing on the top combinations of MFCC coefficient values and DWT levels for both training and validation phases.

For the TESS dataset, utilizing 40 MFCC coefficients combined with DWT level 2 yields strong performance during training, with an accuracy of 99.59% and balanced precision, recall, and F1-score metrics. During validation, the model maintains high accuracy at 96.96% and consistent precision, recall, and F1-score metrics.

In the CREMA dataset, employing 40 MFCC coefficients and DWT level 4 leads to the highest performance. The model achieves an accuracy of 82.86% during training and maintains balanced precision, recall, and F1-score metrics during validation, with an accuracy of 82.80%.

For the SAVEE dataset, utilizing 16 MFCC coefficients paired with DWT level 1 results in the top performance. The model achieves an accuracy of 84.89% during training, with balanced precision, recall, and F1-score metrics. During validation, the model maintains strong performance with an accuracy of 62.50% and consistent precision, recall, and F1-score metrics.

In the RAVDESS dataset, employing 40 MFCC coefficients and DWT level 3 yields the highest performance. The model achieves an accuracy of 83.42% during training and maintains balanced precision, recall, and F1-score metrics during validation, with an accuracy of 42.70%.

Table 5.2.1 Performance of LWCNN-LSTM metrics on each dataset

DATASET	MFCC	DWT	TRAINING				VALIDATION			
			Acc (%)	Pre (%)	Rec (%)	F1S (%)	Acc (%)	Pre (%)	Rec (%)	F1Sc (%)
TESS	40	4	100	100	100	100	99.64	99.64	99.64	99.64
CREMA	40	4	89.08	89.13	89.08	89.06	48.08	47.72	48.08	47.58
SAVEE	40	5	90.36	91.35	90.36	90.34	69.79	69.16	69.79	67.58
RAVDESS	24	2	89.23	89.54	89.23	89.23	52.08	55.63	52.08	52.34

Table 5.2.1 summarizes the top-performing combinations of MFCC coefficient values and DWT levels for each dataset, focusing on the performance metrics of LWCNN-LSTM models during both training and validation phases.

For the TESS dataset, employing 40 MFCC coefficients and DWT level 1 results in exceptional performance, achieving 100% in accuracy, precision, recall, and F1-score during training. During validation, the model maintains high performance with an accuracy of 99.64% and consistent precision, recall, and F1-score metrics.

In CREMA dataset, utilizing 40 MFCC coefficients combined with DWT level 4 yields the highest performance. The model achieves an accuracy of 89.08% during training and maintains comparable metrics during validation, with an accuracy of 89.06% and balanced precision, recall, and F1-score metrics.

For SAVEE dataset, employing 40 MFCC coefficients and DWT level 5 results in the top performance. The model achieves an accuracy of 90.36% during training, with consistent precision, recall, and F1-score metrics. During validation, the model maintains strong performance with an accuracy of 69.79% and balanced precision, recall, and F1-score metrics.

In RAVDESS dataset, utilizing 24 MFCC coefficients paired with DWT level 2 leads to the top performance. The model achieves an accuracy of 89.23% during training and maintains comparable metrics during validation, with an accuracy of 52.08% and balanced precision, recall, and F1-score metrics.

Table 5.2.3 Performance of Resnet18-LSTM metrics on each dataset

DATASET	MFCC	DWT	TRAINING				VALIDATION			
			Acc (%)	Pre (%)	Rec (%)	F1S (%)	Acc (%)	Pre (%)	Rec (%)	F1S (%)
TESS	40	4	100	100	100	100	99.82	99.82	99.82	99.82
CREMA	32	2	87.42	87.51	87.42	87.39	51.37	50.76	51.37	50.59
SAVEE	24	1	100	100	100	100	80.20	82.83	80.20	80.47
RAVDESS	32	2	100	100	100	100	68.40	68.80	68.40	68.23

Table 5.2.3 summarizes the top-performing combinations of MFCC coefficient values and DWT levels for each dataset, focusing on the performance metrics of Resnet18-LSTM models during both training and validation phases.

TESS Dataset: Using 40 MFCC coefficients and DWT level 4, the model achieved perfect accuracy, precision, recall, and F1-score during training. In the validation phase, it maintained exceptionally high performance with an accuracy of 99.82% and consistent precision, recall, and F1-score metrics.

CREMA Dataset: Employing 32 MFCC coefficients and DWT level 2 resulted in an accuracy of 87.42% during training. However, during validation, the model's performance decreased, with an accuracy of 51.37% and lower precision, recall, and F1-score metrics.

SAVEE Dataset: Utilizing 24 MFCC coefficients paired with DWT level 1 led to perfect performance during training, with 100% accuracy, precision, recall, and F1-score. In the validation phase, the model maintained strong performance with an accuracy of 80.20% and balanced precision, recall, and F1-score metrics.

RAVDESS Dataset: Using 32 MFCC coefficients and DWT level 2 resulted in perfect performance during training, similar to the SAVEE dataset. However, during validation, the model's accuracy decreased to 68.40%, with slightly lower precision, recall, and F1-score metrics.

Overall, the ResNet18-LSTM model demonstrates impressive performance across datasets, especially in terms of accuracy during training. However, its performance varies during validation, with notable drops in accuracy and other metrics for some datasets like CREMA and RAVDESS.

The ResNet18-LSTM model emerges as the standout performer among the three models

considered—LSTM, LWCNN-LSTM, and ResNet18-LSTM—across the diverse speech emotion recognition datasets of CREMA, RAVDESS, SAVEE, and TESS. With consistently superior performance, particularly highlighted by its near-perfect accuracy on the TESS dataset and robust performance across other datasets, ResNet18-LSTM establishes itself as a formidable choice for this task. Its ability to effectively capture complex temporal and spatial features inherent in emotional speech data underscores its potential to advance speech emotion recognition technology. While minor fluctuations in accuracy during validation on select datasets were observed, the overall strength and reliability of the ResNet18-LSTM model position it as a promising solution for real-world applications requiring nuanced emotion detection from speech signals. Resnet18-LSTM models consistently demonstrate superior accuracy and stability across datasets, particularly evident in the TESS and SAVEE datasets. The top combinations of MFCC coefficients and DWT levels for Resnet18-LSTM models often yield higher accuracy and robustness compared to LSTM models, indicating the effectiveness of leveraging both convolutional and recurrent neural network architectures for speech emotion recognition tasks. However, LSTM models also exhibit competitive performance, especially in datasets such as CREMA and RAVDESS, where specific combinations of MFCC coefficients and DWT levels lead to optimal results. These findings underscore the versatility and efficacy of Resnet18-LSTM and LWCNN-LSTM architectures in capturing complex temporal and spatial features inherent in emotional speech data. Both models achieve higher Accuracy, Precision, Recall, and F1-Score compared to LSTM models on most datasets and emotions in the table. This suggests that the Resnet18-LSTM architecture may be more effective for capturing the features relevant to emotion recognition in speech.

5.2.2 POSSIBLE REASONS FOR RESNET18 and LWCNN-LSTM'S PERFORMANCE

The results in the Table 5.1, 5.2 and 5.3 highlight a noteworthy observation: All Resnet18-LSTM LWCNN-LSTM and LSTM models achieve considerably higher Accuracy, Precision, Recall, and F1-Score on the TESS dataset compared to the other benchmark datasets (CREMA-D, RAVDESS, SAVEE).

Convolutional Neural Networks (CNNs) are adept at capturing spatial features from data. In speech emotion recognition, CNNs can effectively learn patterns from the spectrogram, a visual representation of the speech signal that encodes frequency information over time. This can be particularly useful for identifying emotions that are influenced by vocal characteristics like pitch variations. LSTMs are powerful at capturing sequential features. They can learn long-

term dependencies within speech signals, which is important for understanding the emotional context of spoken language. However, for speech emotion recognition, the combination of CNNs for spatial features and LSTMs for sequential features might be particularly effective.

The ResNet18 architecture, with its deep convolutional neural network design and residual connections, excels in feature extraction from complex data such as speech signals, allowing the model to effectively capture important spatial features. Additionally, the integration of LSTM layers enables the model-to-model temporal dependencies and capture long-term relationships within the data, which is crucial for understanding the nuanced dynamics of emotional speech. The combination of these architectures likely results in a more comprehensive representation of the data, facilitating more accurate emotion recognition.

5.2.3 PERFORMANCE VARIATIONS ACROSS DATASETS

5.2.3.1 LSTM CURVES

A. Training

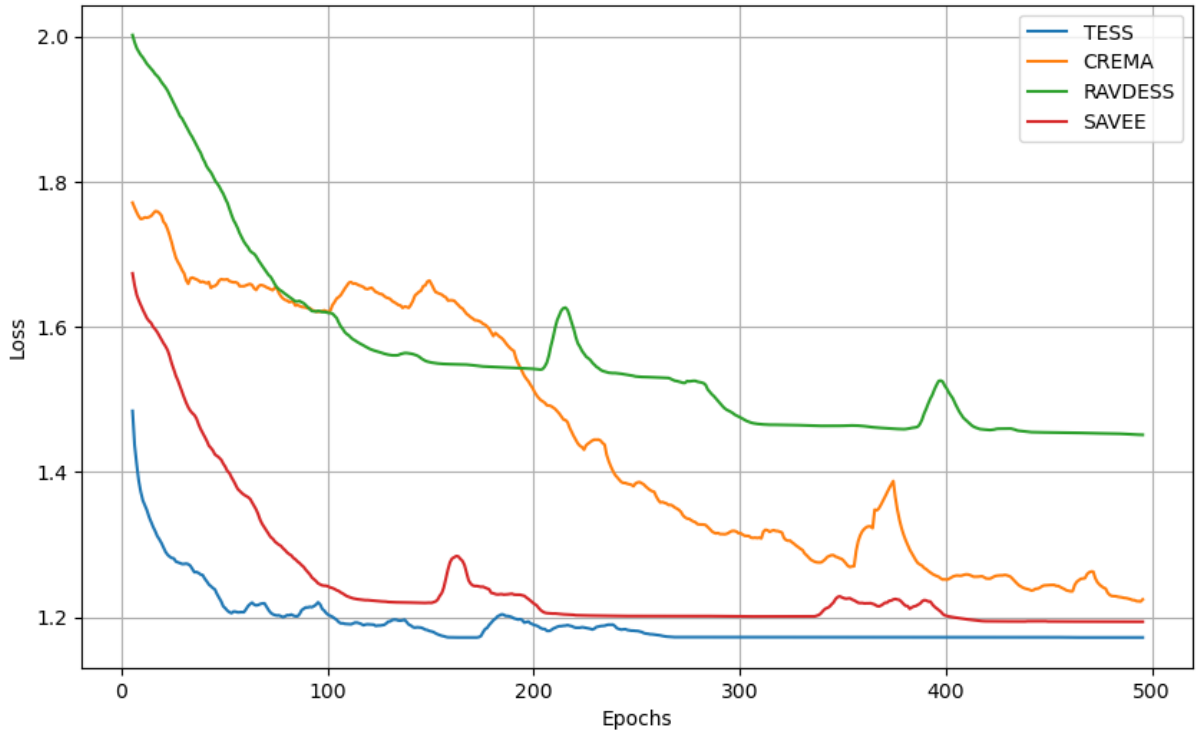


Fig 5.3.1 Training Loss curve for LSTM model.

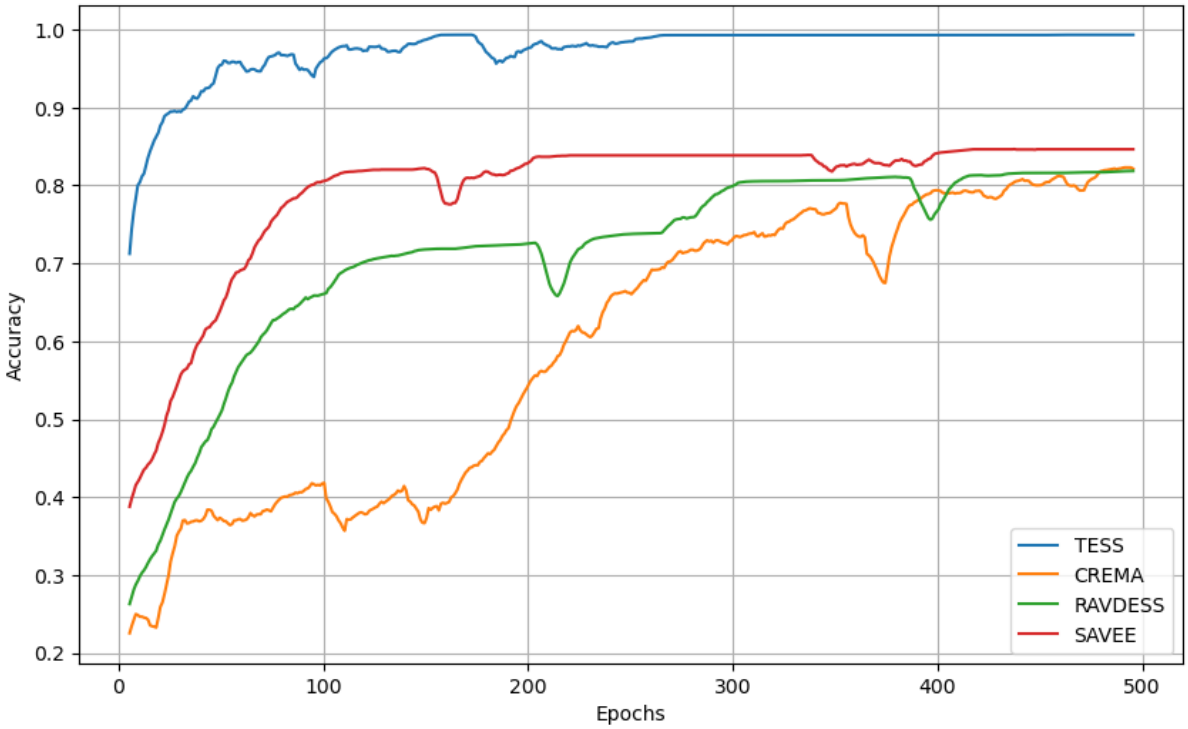


Fig 5.3.2 Training accuracy curve for LSTM model.

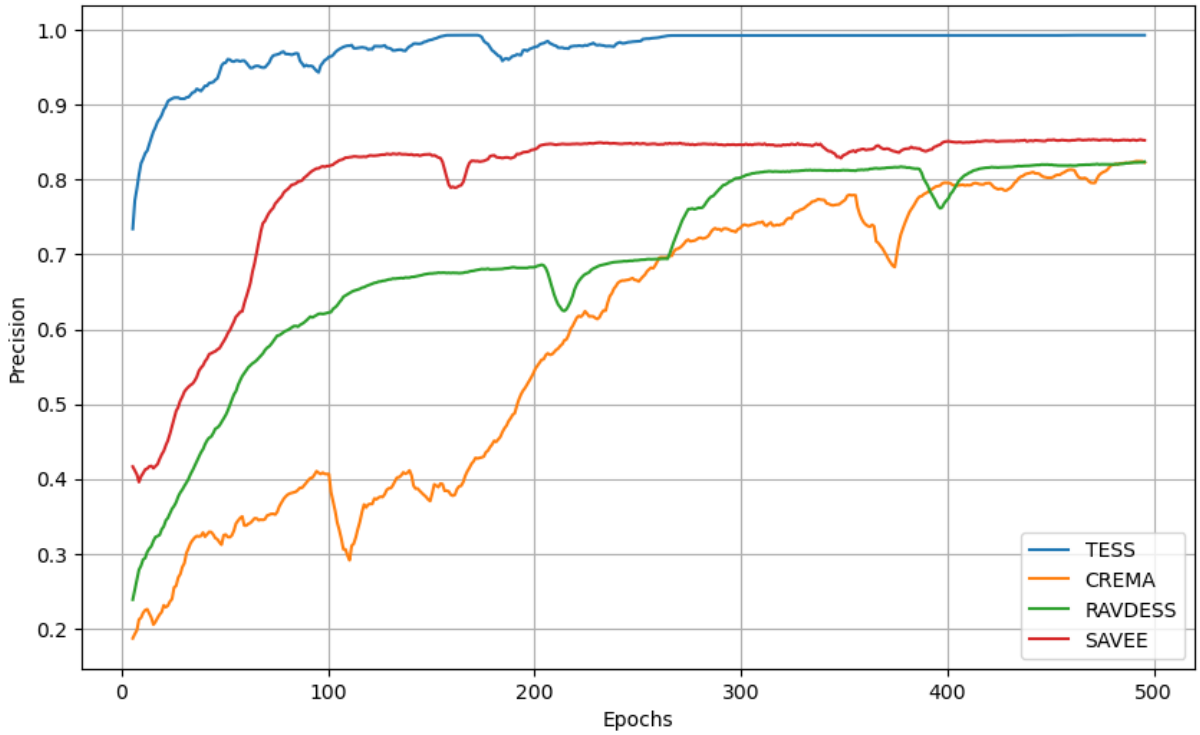


Fig 5.3.3 Training precision curve for LSTM model.

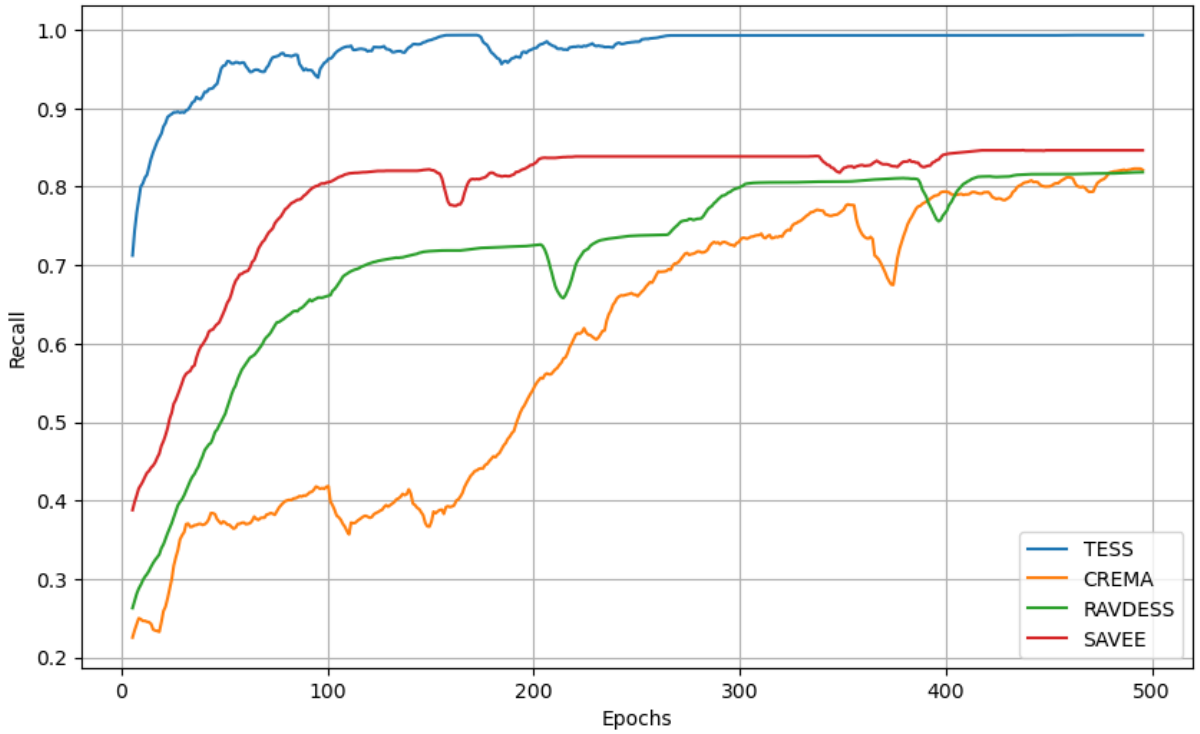


Fig 5.3.4 Training recall curve for LSTM model.

B. Validation

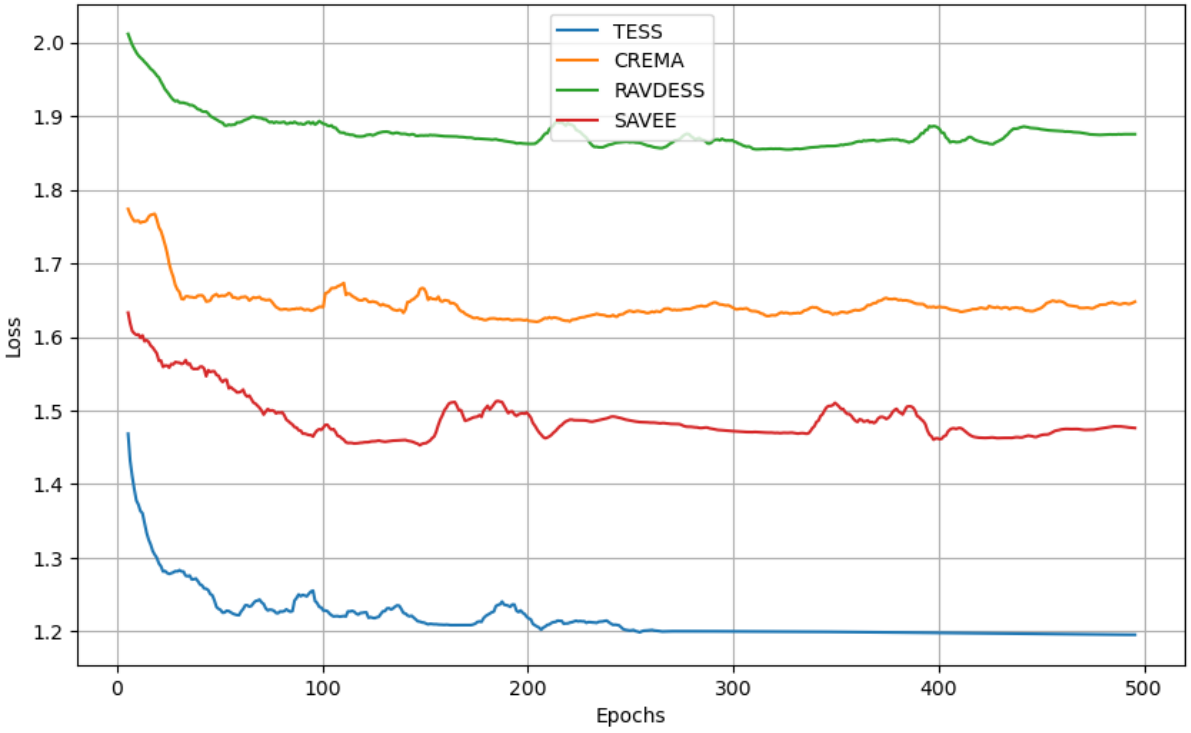


Fig 5.3.5 Validation loss curve for LSTM model

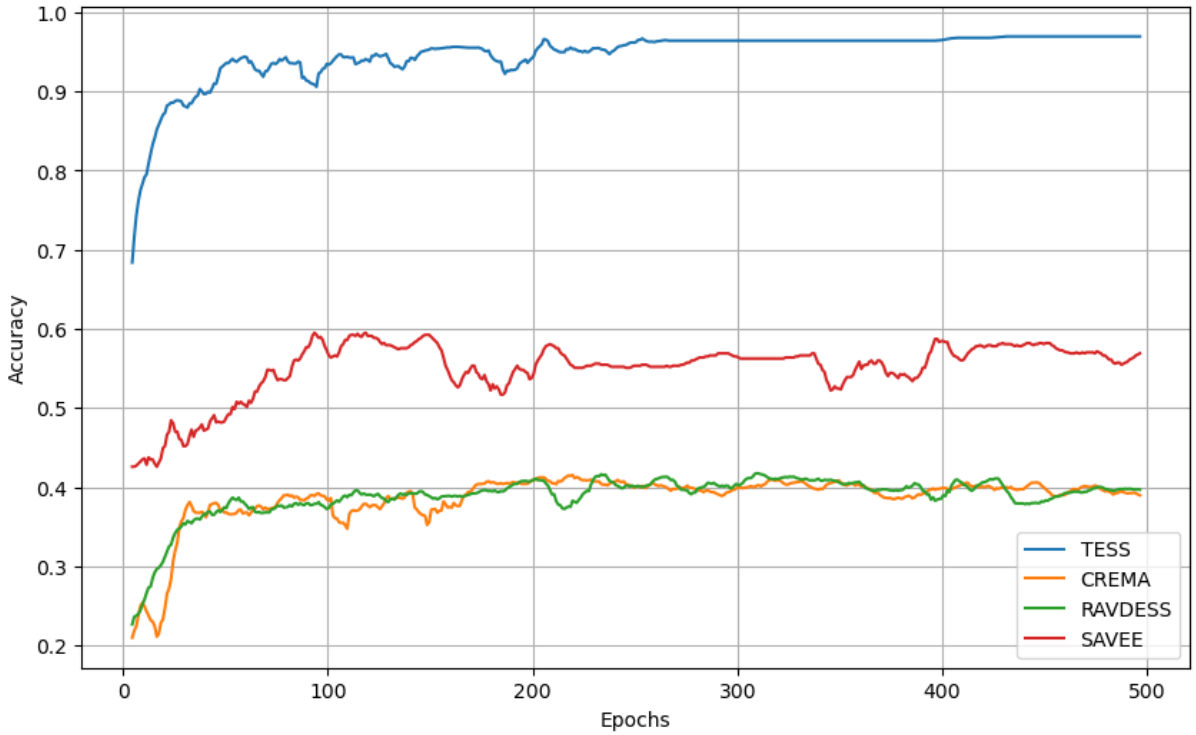


Fig 5.3.6 Validation accuracy curve for LSTM model

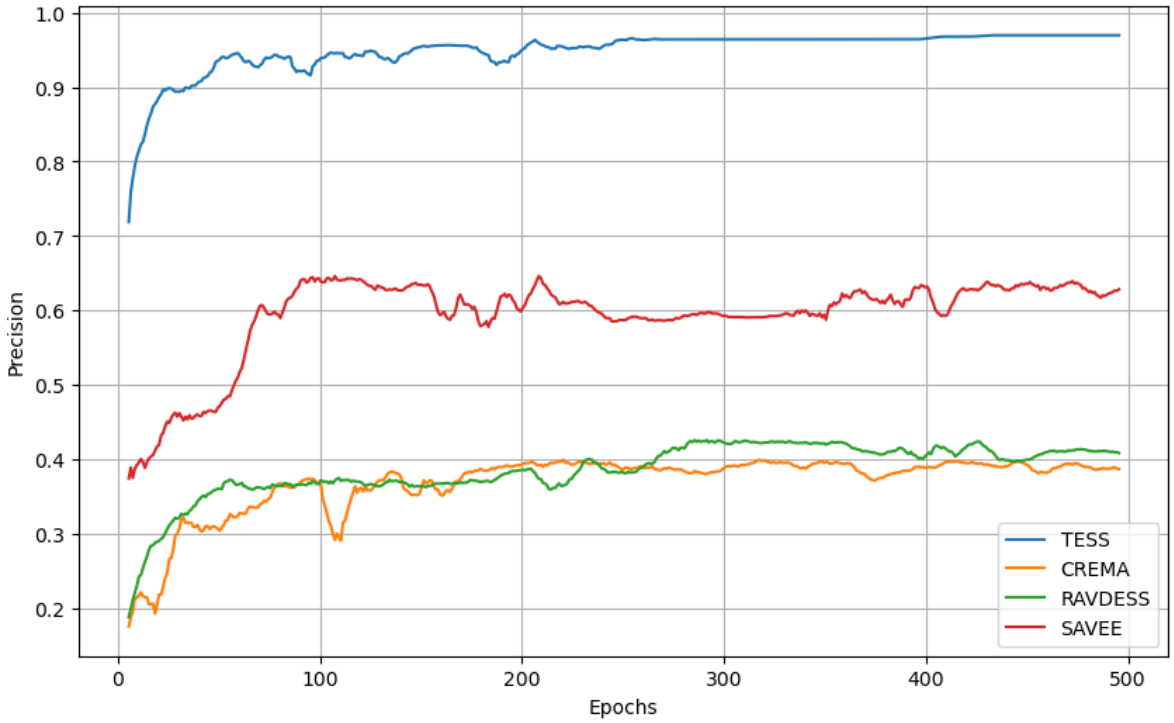


Fig 5.3.7 Validation precision curve for LSTM model

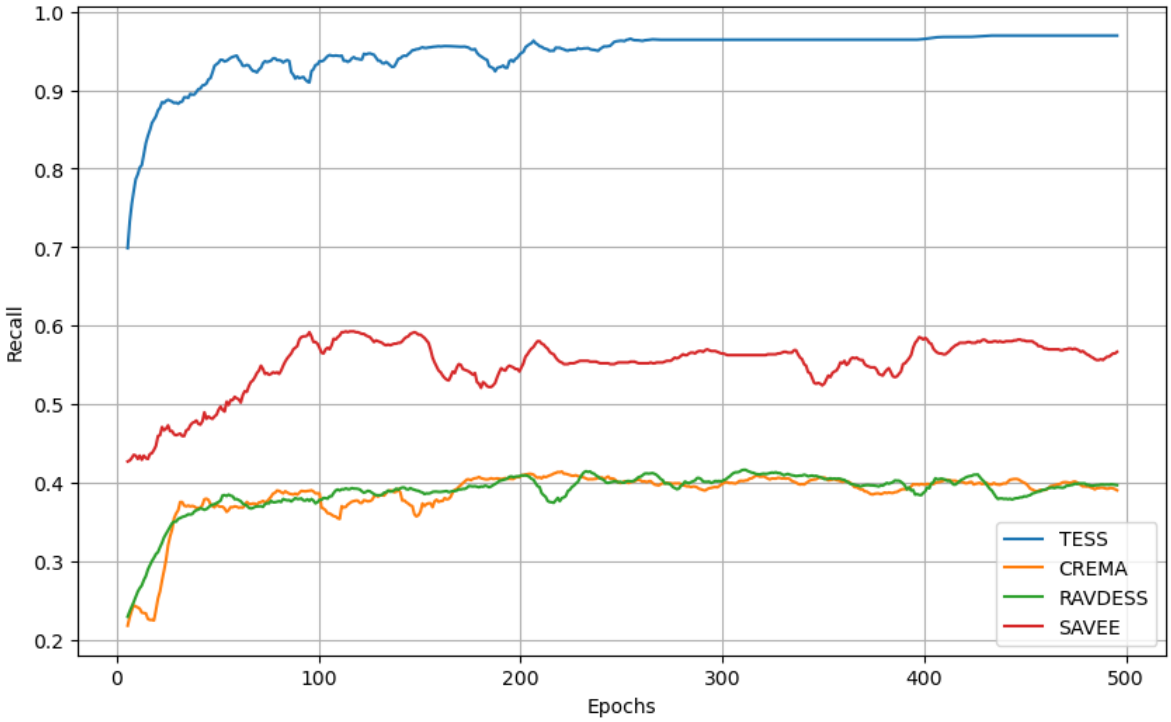


Fig 5.3.8 Validation recall curve for LSTM model

The training loss (Figure 5.3.1) and validation loss (Figure 5.3.5) for all four datasets follows a downward trend over epochs, which suggests that all four LSTM models are successfully learning from the data. The loss values decrease over epochs, indicating improved performance. However, there are fluctuations and plateaus in loss values, suggesting periods of slower learning or convergence. The TESS dataset appears to have the lowest training loss throughout the epochs. This suggests that the LSTM model trained on the TESS dataset achieved the lowest loss among the four.

The training and validation curves (Figure 5.3.2 - 5.3.4, 5.3.6 - 5.3.8) for all four datasets follows an upward trend over epochs, which suggests that all four LSTM models are improving their ability to generalize to unseen data as training progresses. The TESS dataset appears to have the highest validation accuracy throughout the epochs. This suggests that the LSTM model trained on the TESS dataset.

5.3.3.2 LWCNN- LSTM CURVES

A. Training

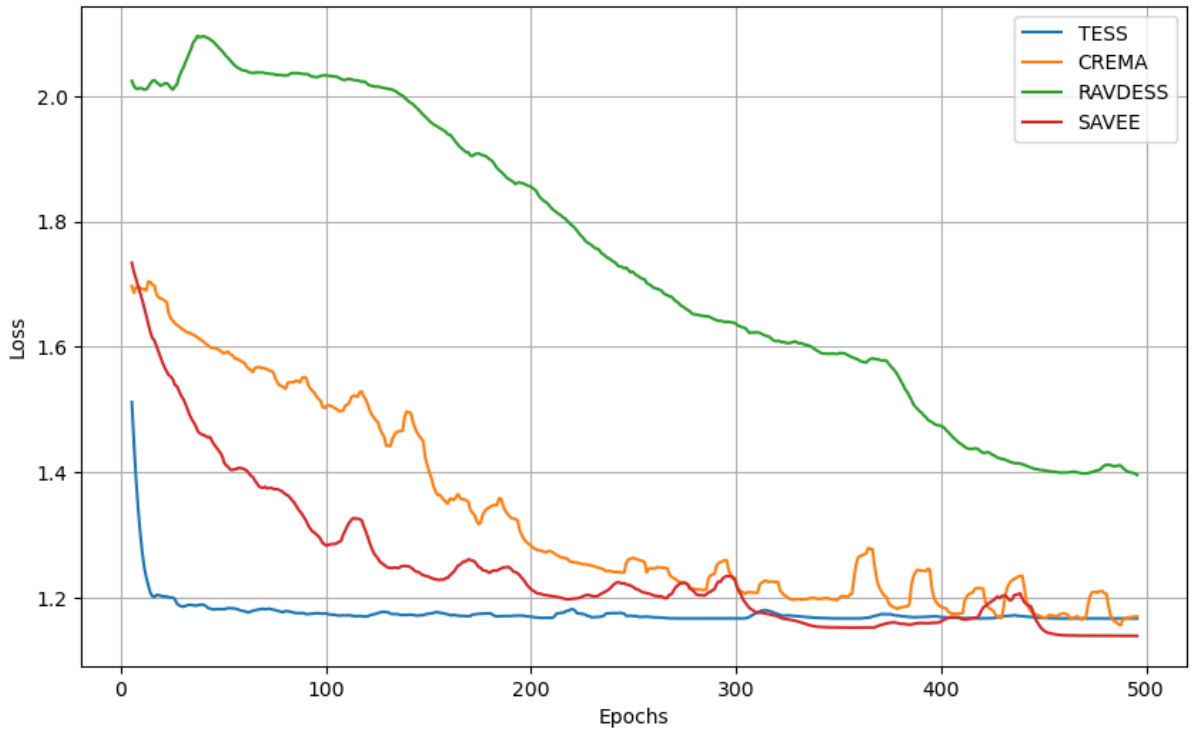


Fig 5.2.9 Training loss curve for LWCNN-LSTM model

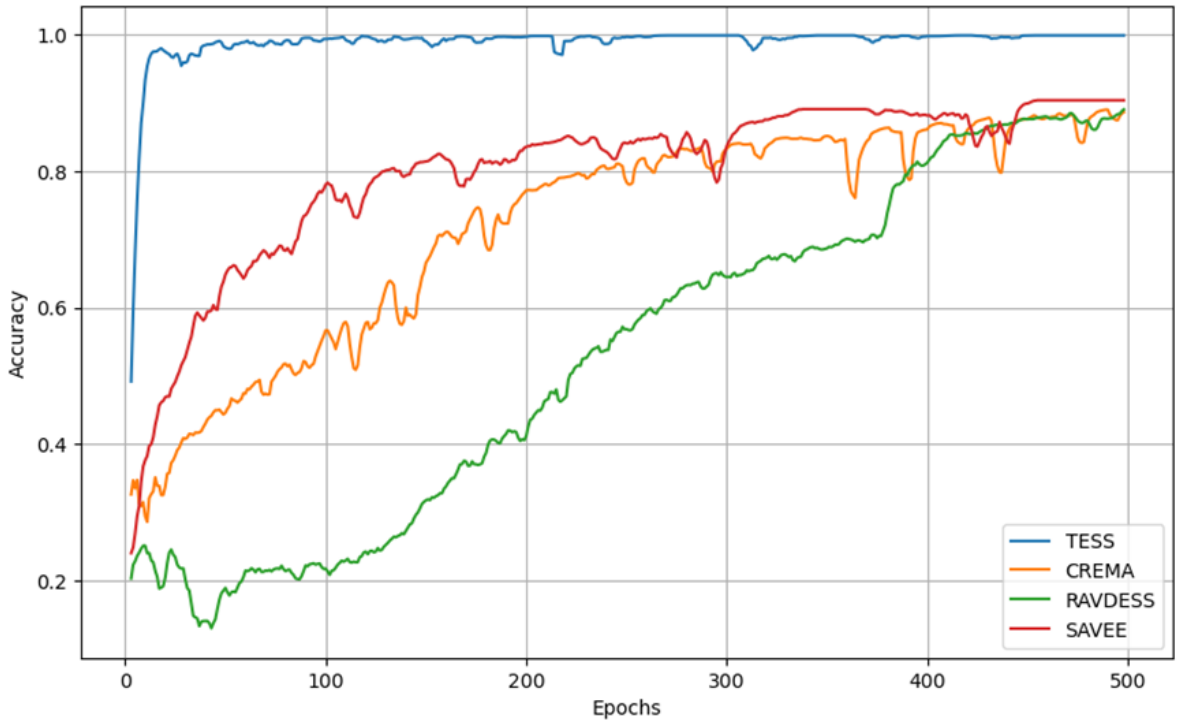


Fig 5.2.10 Training accuracy curve for LWCNN-LSTM model

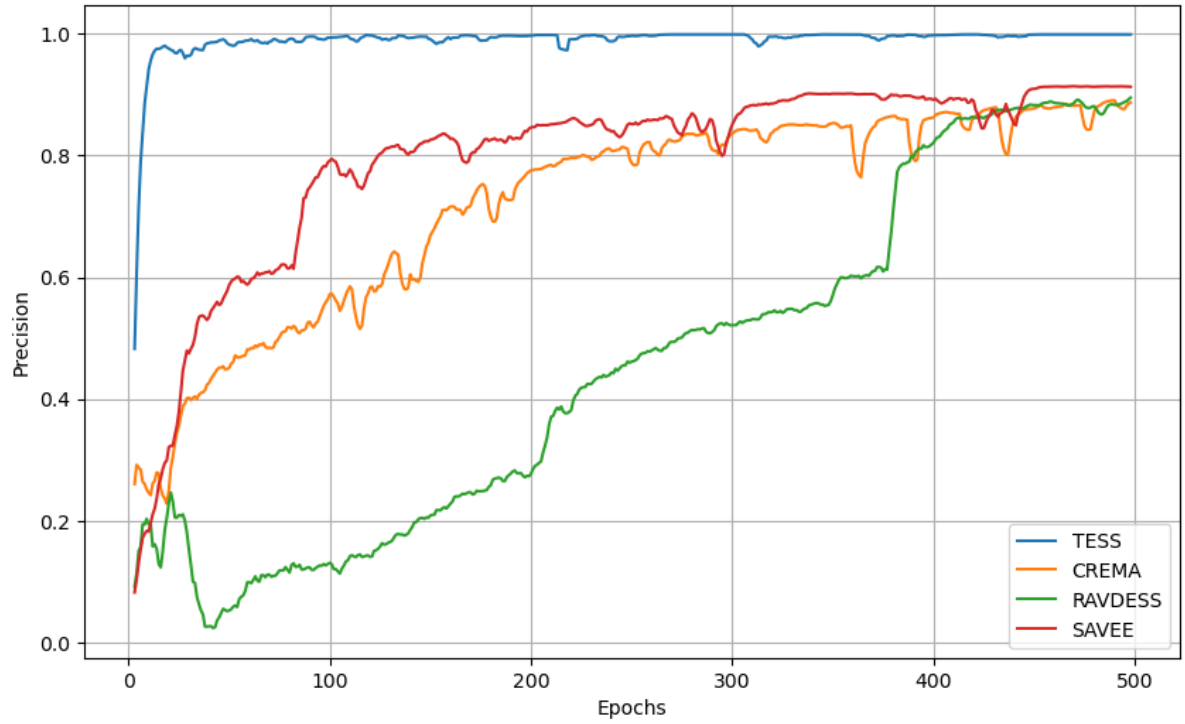


Fig 5.2.11 Training precision curve for LWCNN-LSTM model

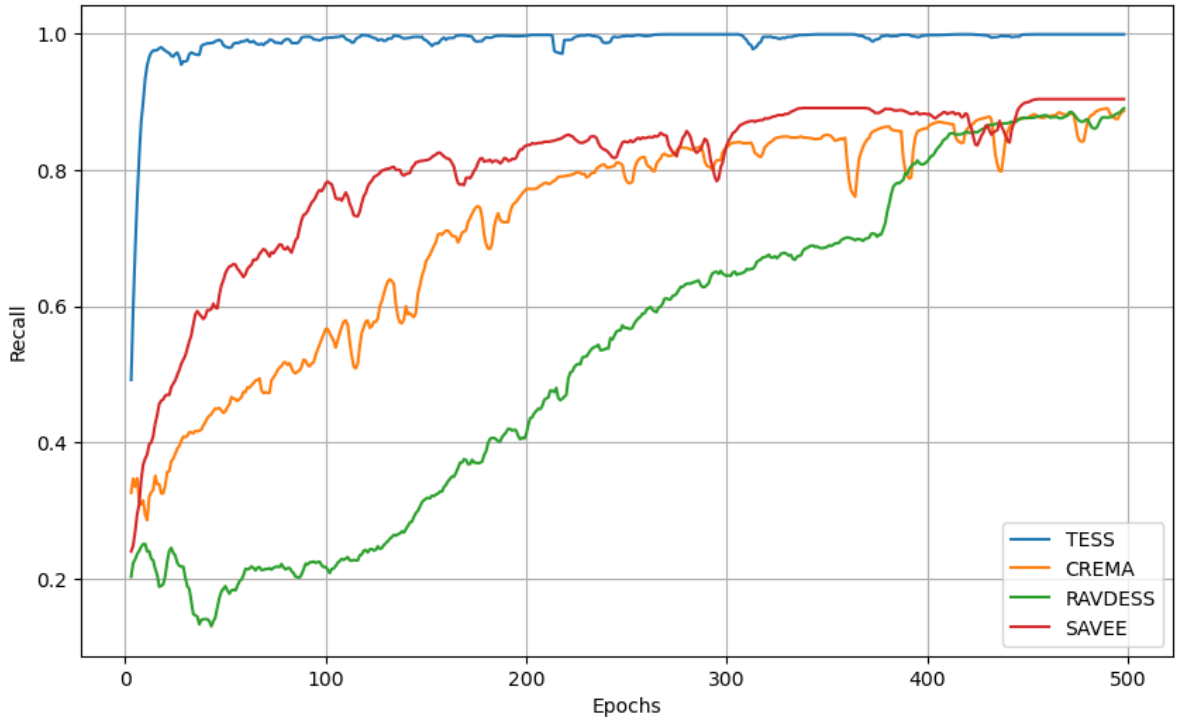


Fig 5.2.12 Training recall curve for LWCNN-LSTM model

B. Validation

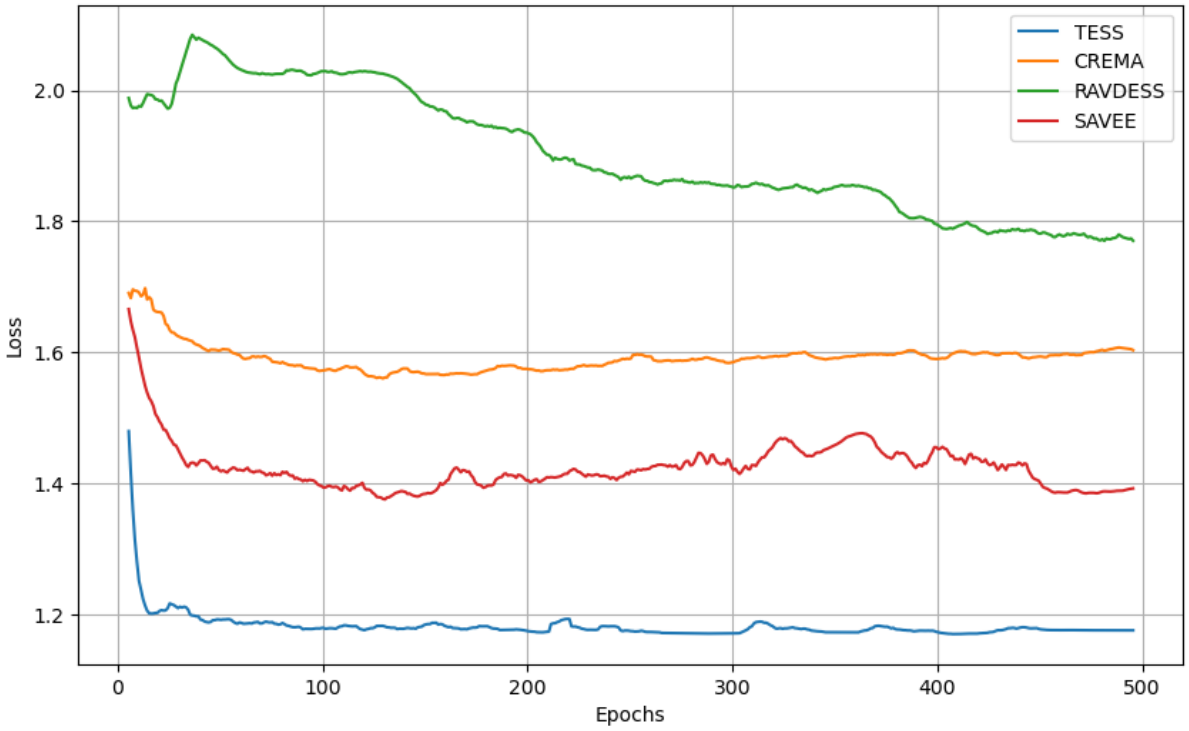


Fig 5.2.13 Validation loss curve for LWCNN-LSTM model

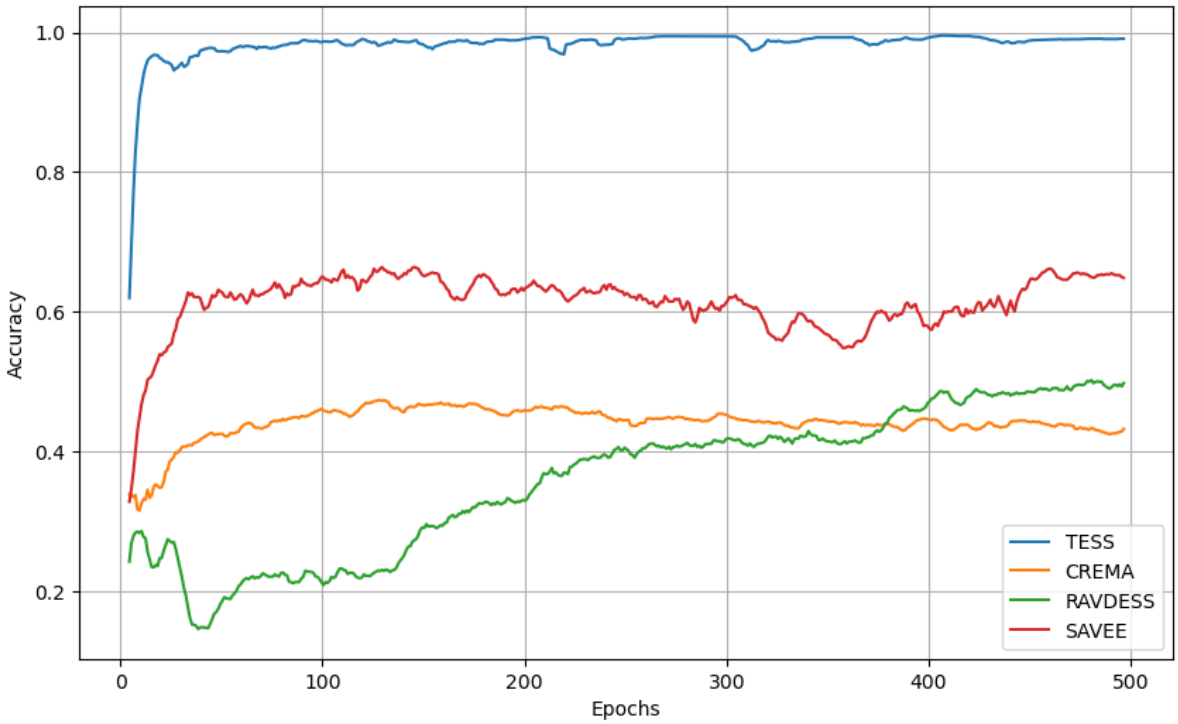


Fig 5.2.14 Validation accuracy curve for LWCNN-LSTM model



Fig 5.2.15 Validation precision curve for LWCNN-LSTM model

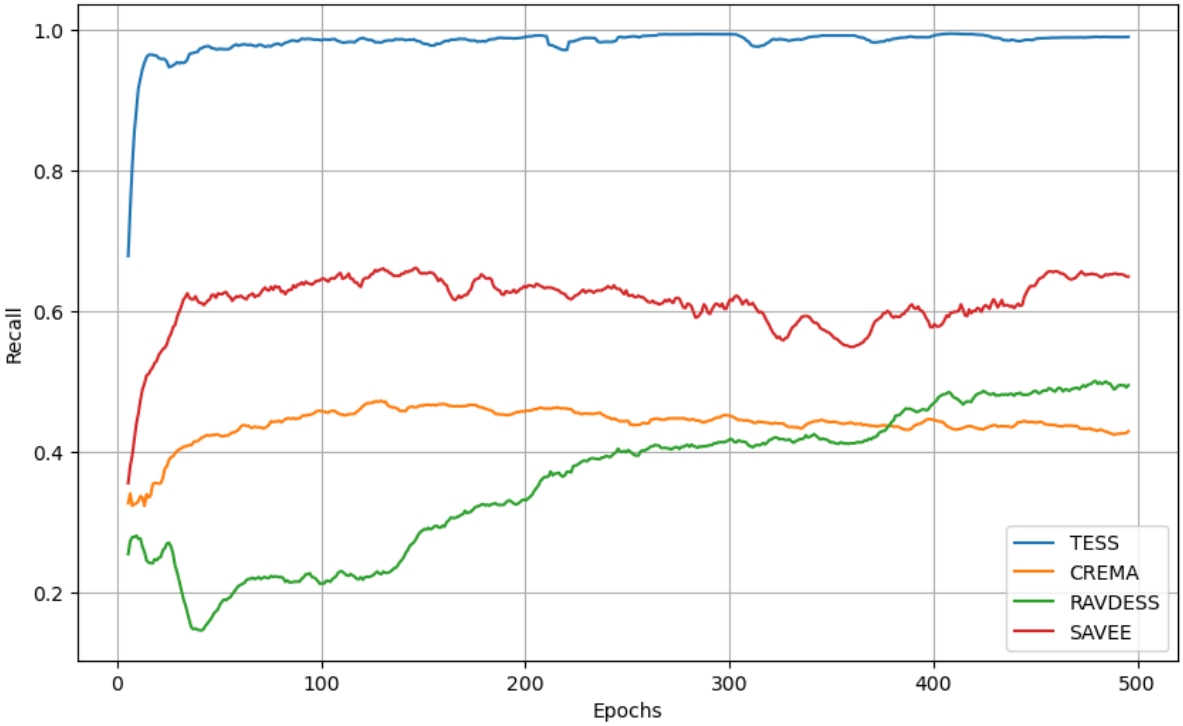


Fig 5.2.16 Validation recall curve for LWCNN-LSTM model

Similar trends in loss and performance curve were observed both in LSTM and LWCNN-LSTM. The LWCNN-LSTM (Figure 5.2.9 - 5.2.16) slightly had an increase in performance and a better loss. The TESS dataset demonstrably led to superior model

performance compared to the other datasets, achieving both higher validation accuracy and a smaller gap between training and validation metrics. Notably, while accuracy demonstrates a consistent upward trend, metrics such as recall and precision exhibit fluctuations, indicating nuanced performance variations across different datasets and classes. These findings suggest a multifaceted learning capability within the model, potentially highlighting both strengths and weaknesses in its classification abilities. The LWCNN-LSTM model demonstrated an overall increase in performance metrics compared to the standalone LSTM model on all four datasets (TESS, CREMA, RAVDESS, SAVEE). This improvement suggests that the CNN component in the LWCNN-LSTM architecture might be effective in capturing local features in the speech spectrograms, while the LSTM handles the temporal dependencies within the sequences.

5.3.3.3 Resnet18-LSTM CURVES

A. Training

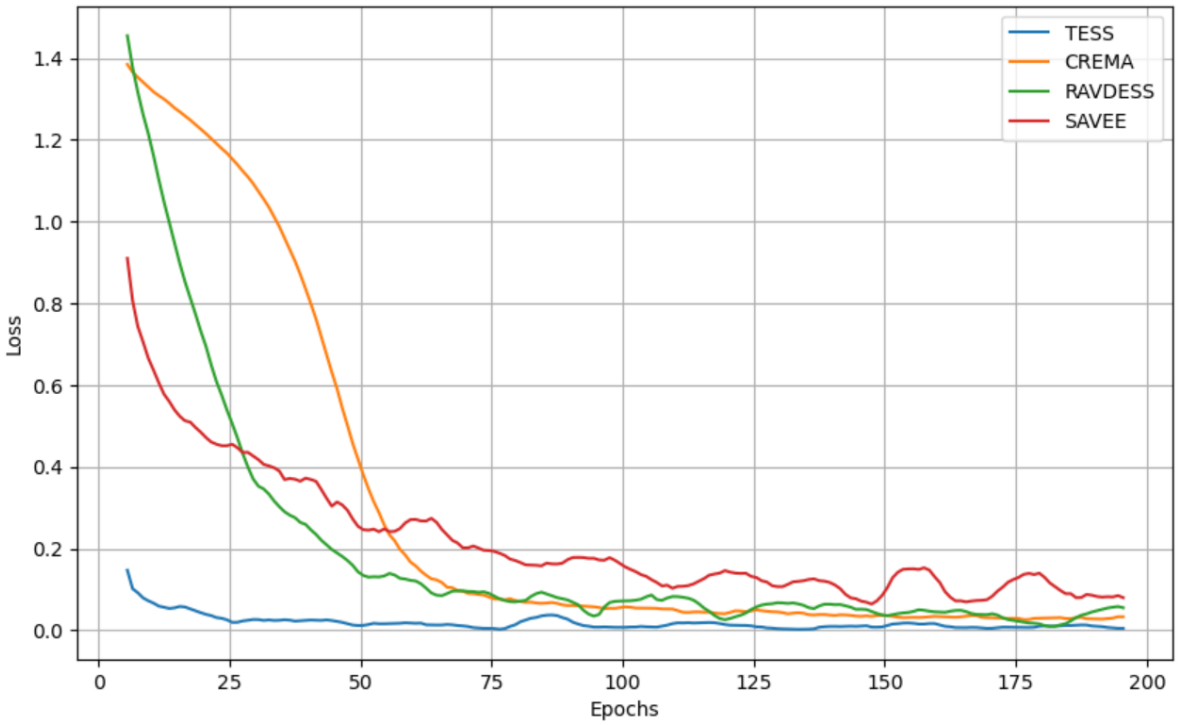


Fig 5.2.17 Training Loss curve for Resnet18-LSTM model

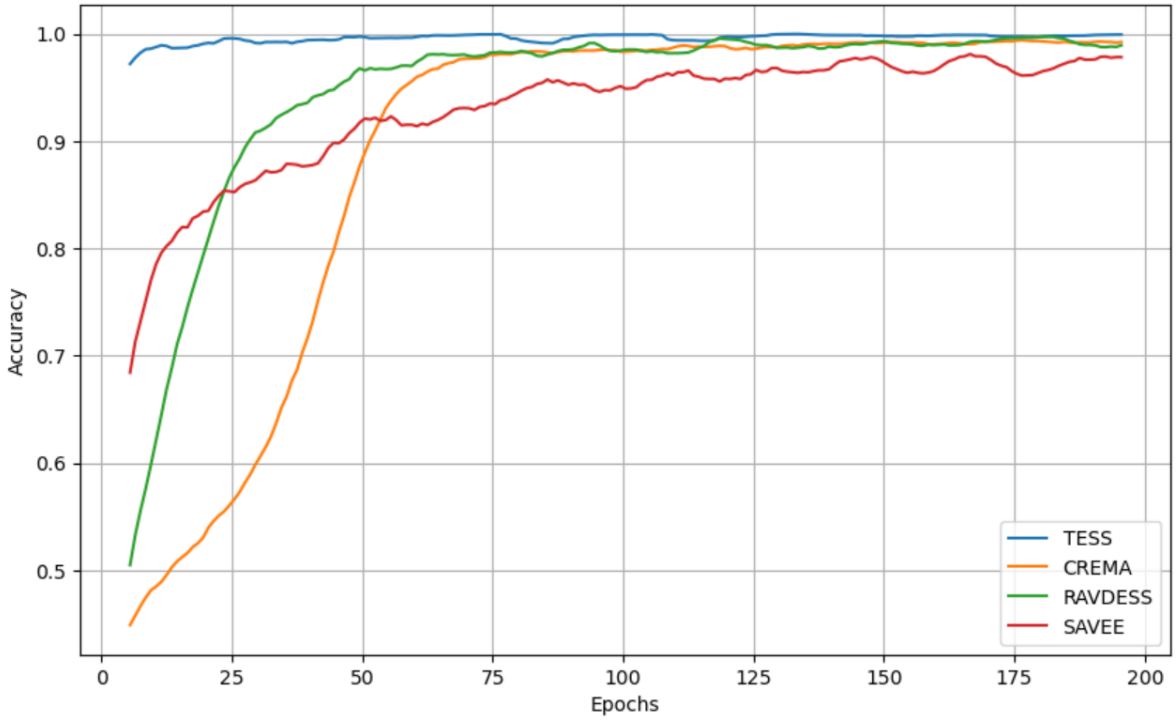


Fig 5.2.18 Training Accuracy curve for Resnet18-LSTM model

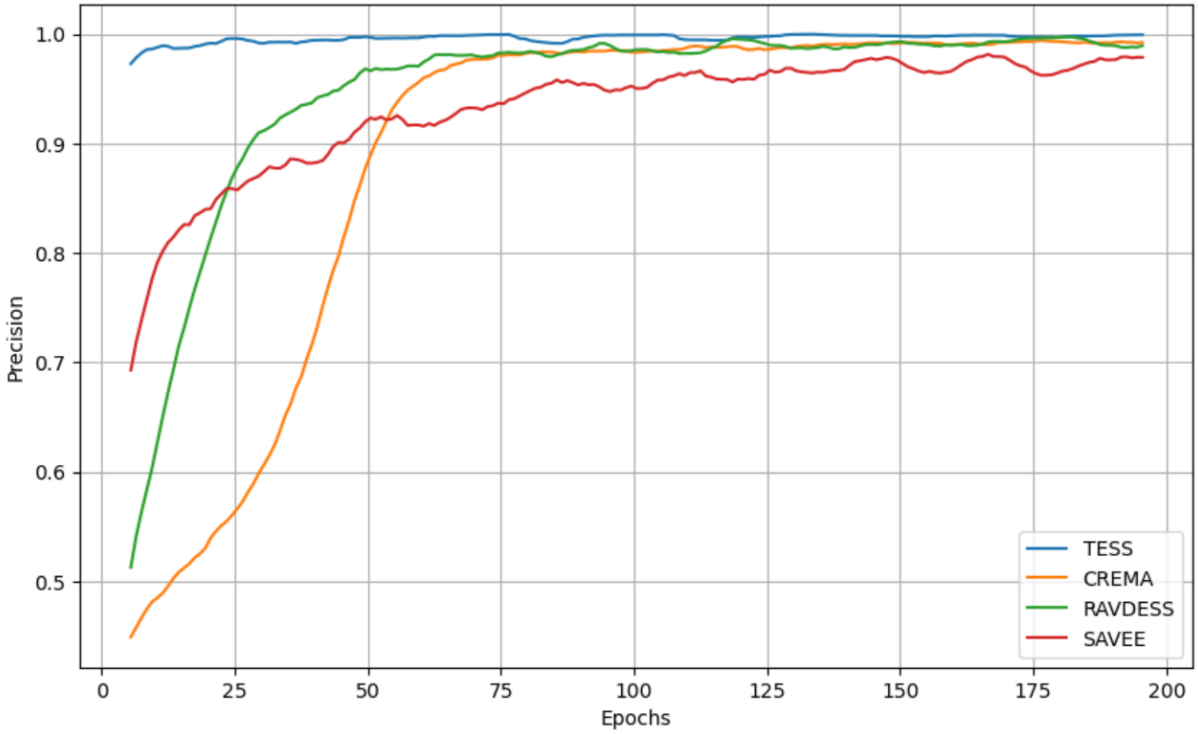


Fig 5.2.19 Training Precision curve for Resnet18-LSTM model

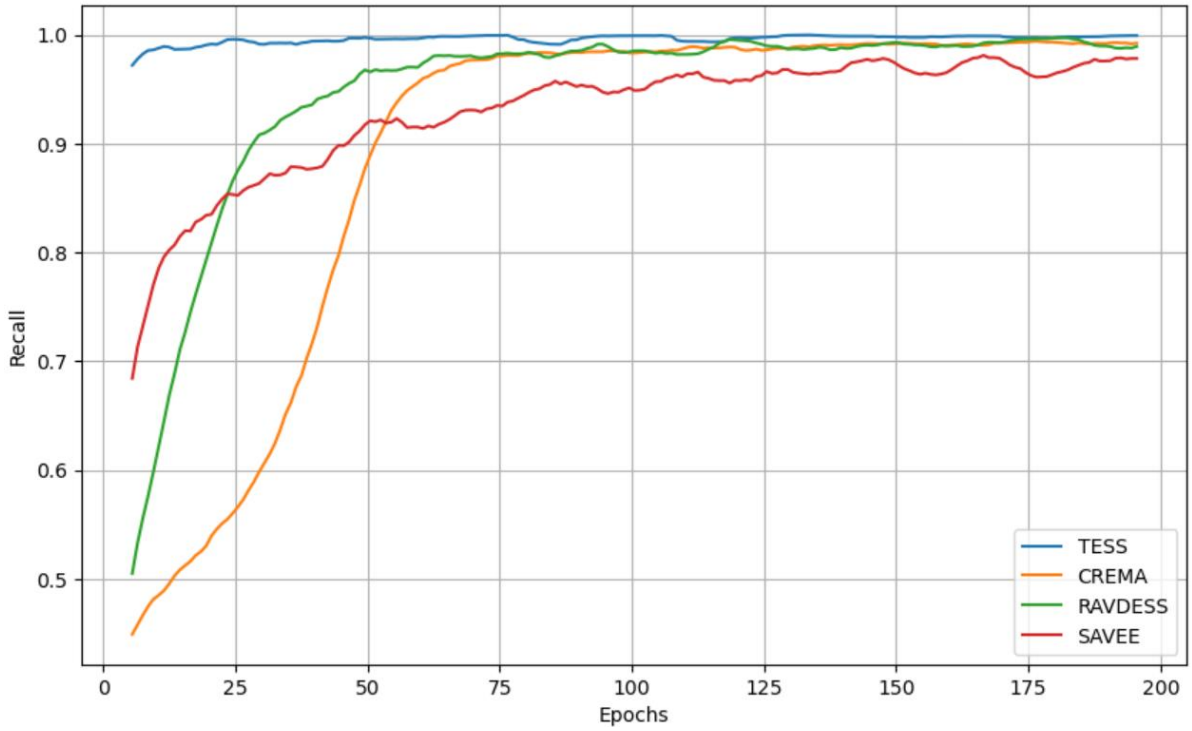


Fig 5.2.20 Training Recall curve for Resnet18-LSTM model

A. Validation

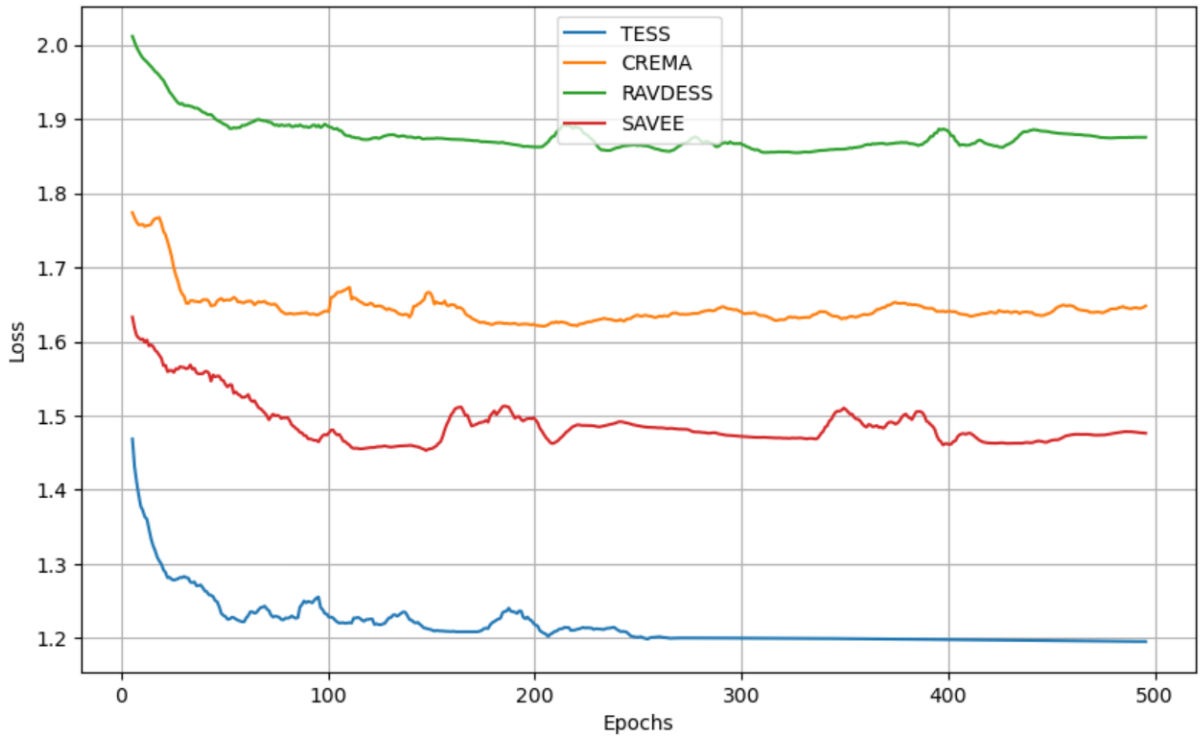


Fig 5.2.21 Validation Loss curve for Resnet18-LSTM model

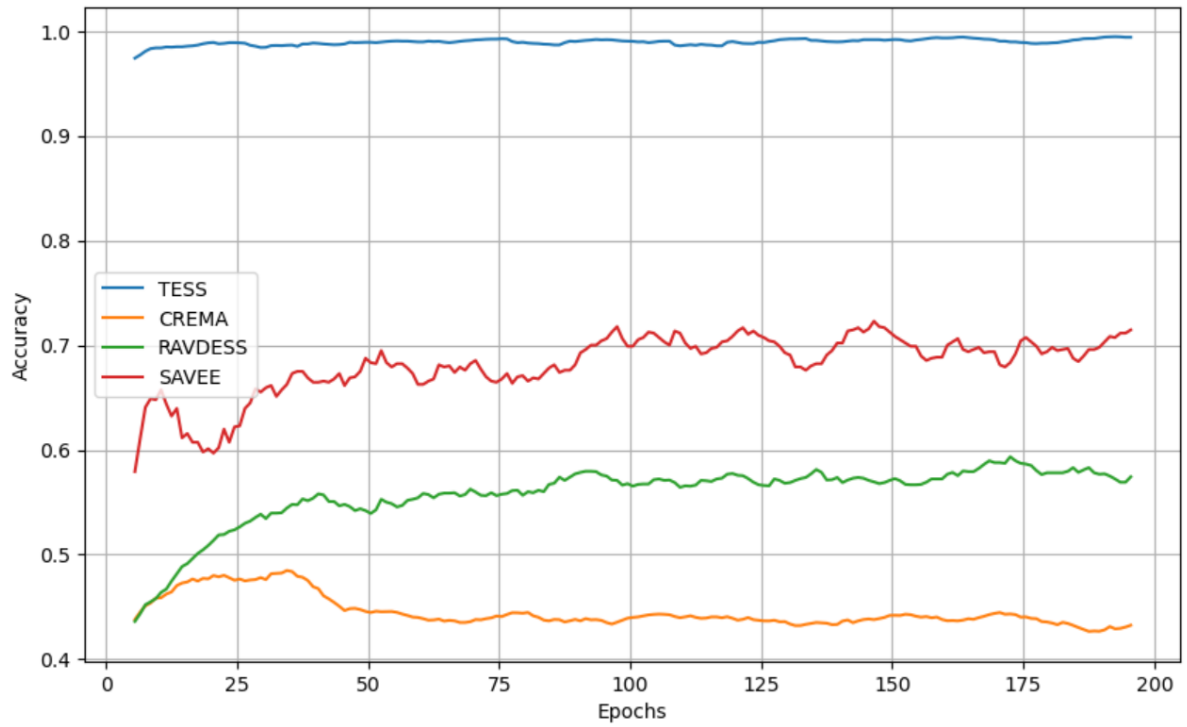


Fig 5.2.22 Validation Accuracy curve for Resnet18-LSTM model

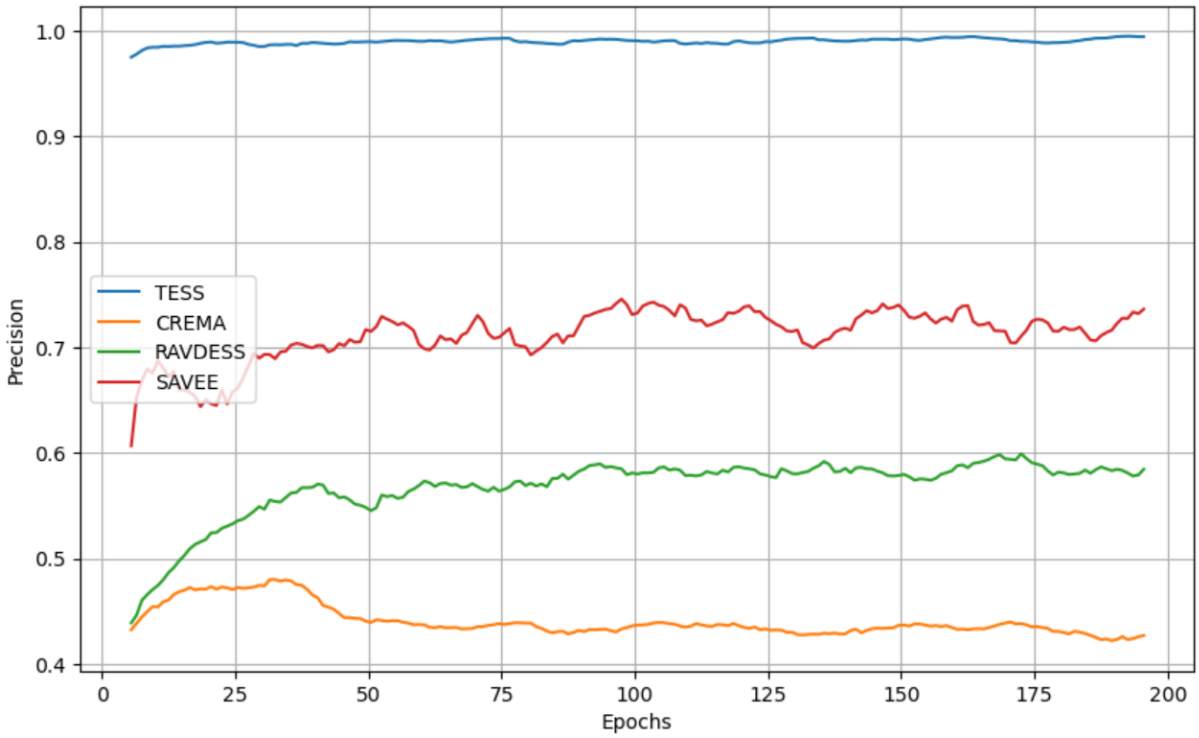


Fig 5.2.23 Validation Precision curve for Resnet18-LSTM model

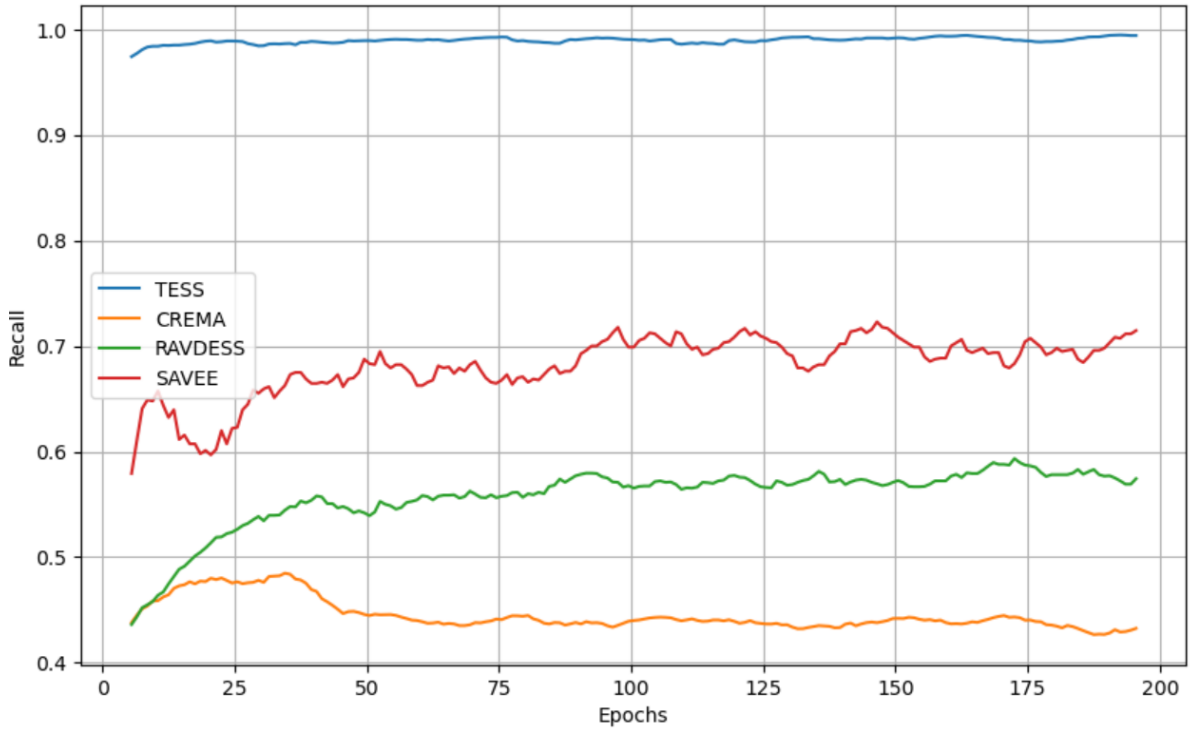


Fig 5.2.24 Validation Recall curve for Resnet18-LSTM model

5.2.4 CONFUSION MATRIX

A. TESS

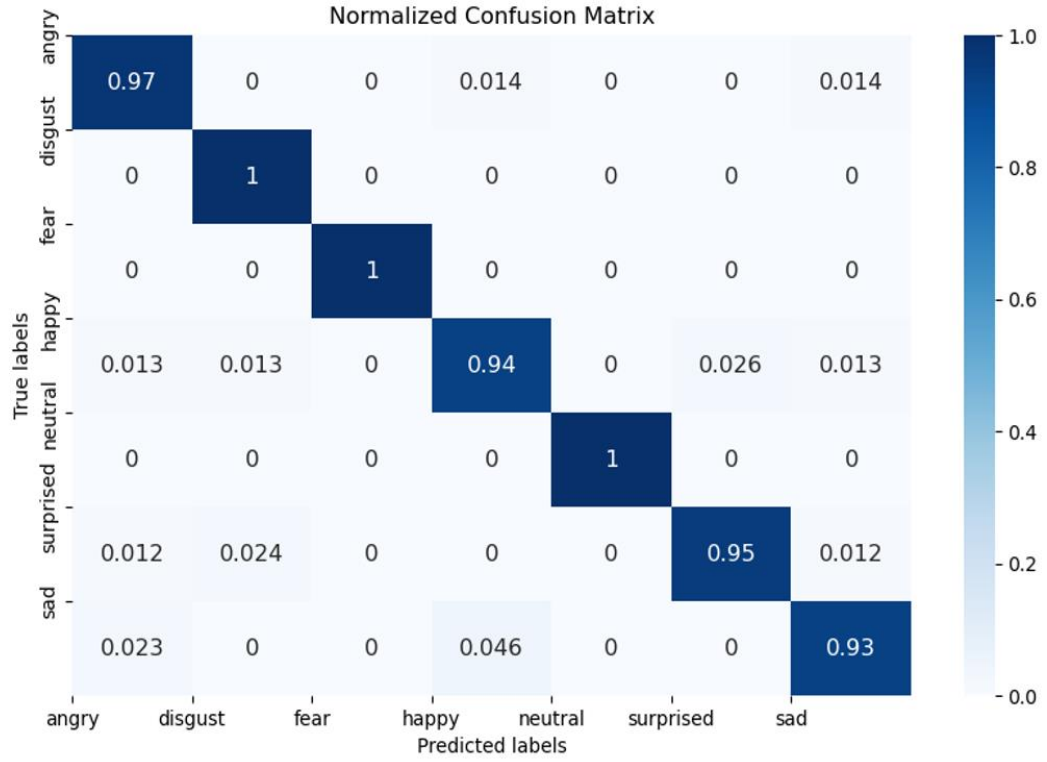


Fig 5.2.4.1 Confusion Matrix for LSTM model - TESS

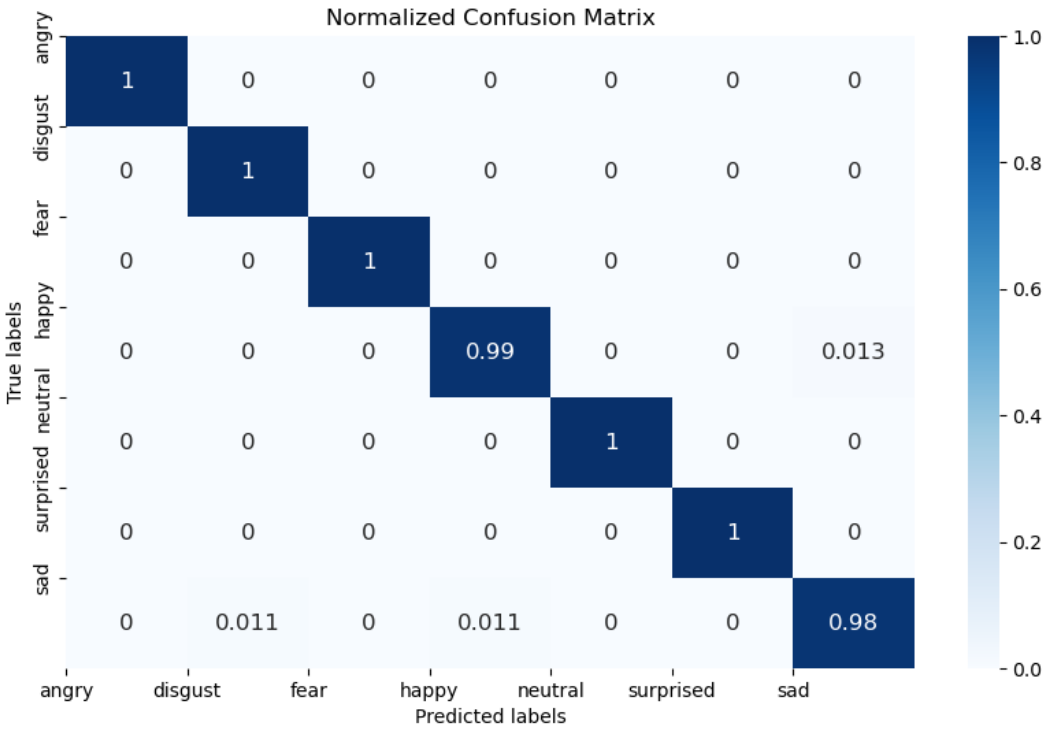


Fig 5.2.4.2 Confusion Matrix for LWCNN-LSTM model – TESS

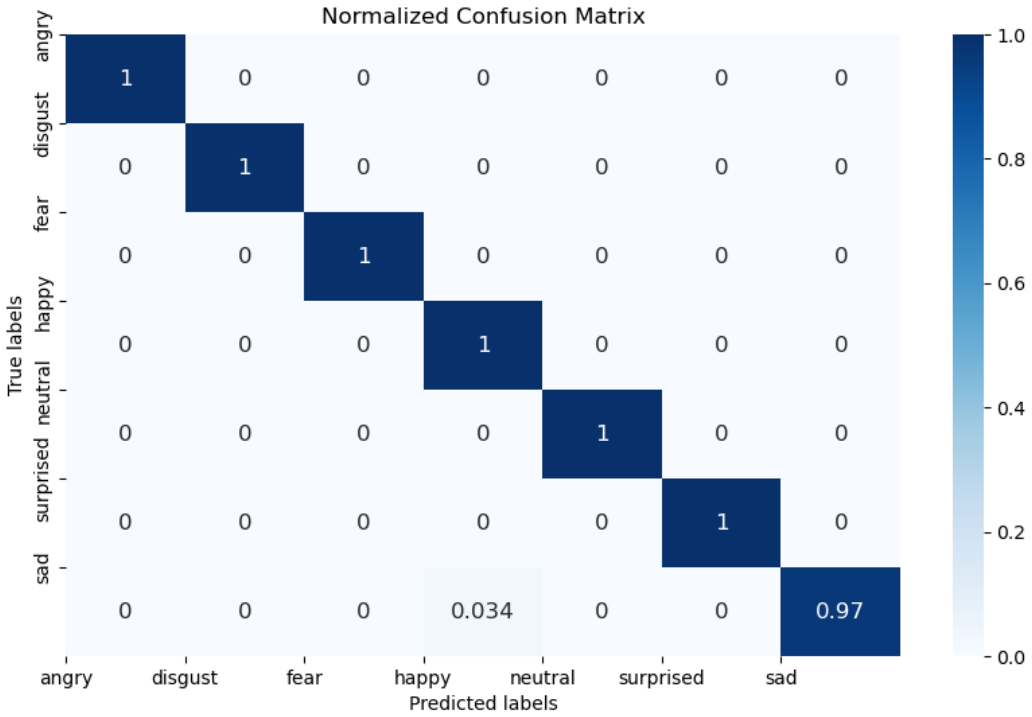


Fig 5.2.4.3 Confusion Matrix for Resnet18-LSTM model – TESS

B. CREMA

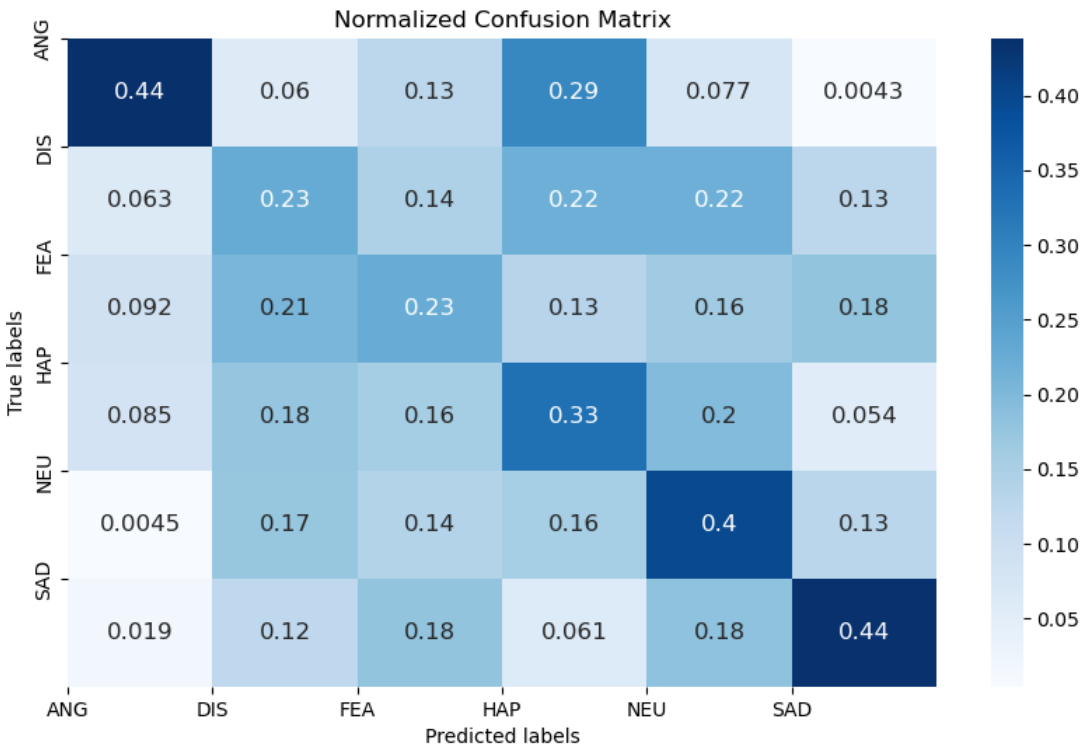


Fig 5.2.4.4 Confusion Matrix for LSTM model - CREMA

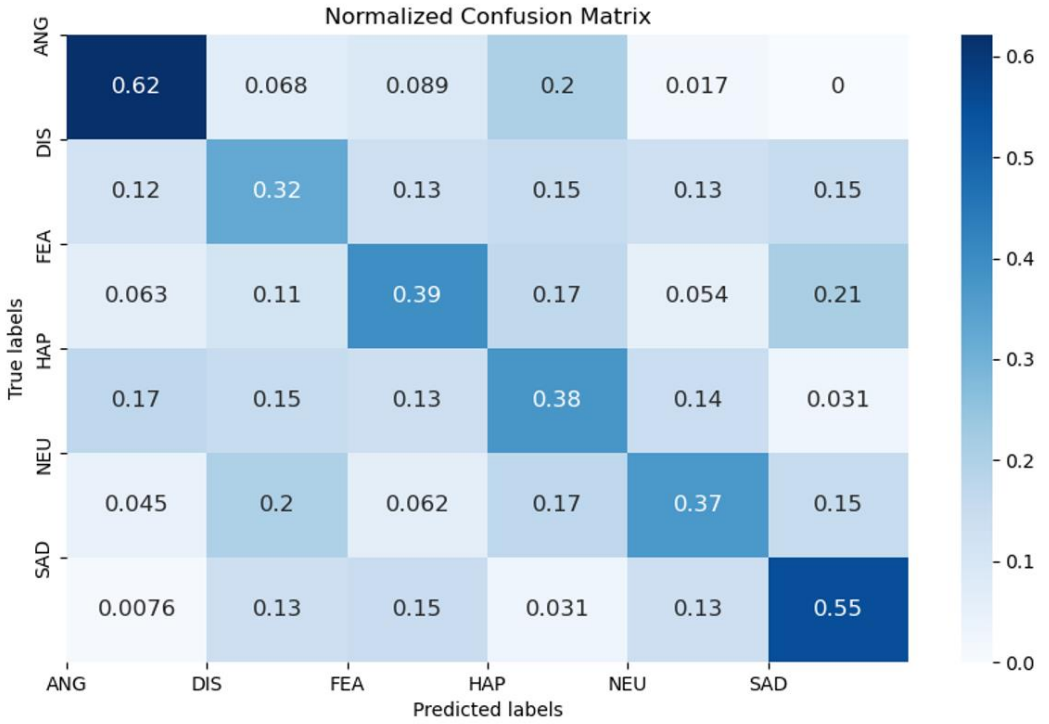


Fig 5.2.4.5 Confusion Matrix for LWCNN-LSTM model – CREMA

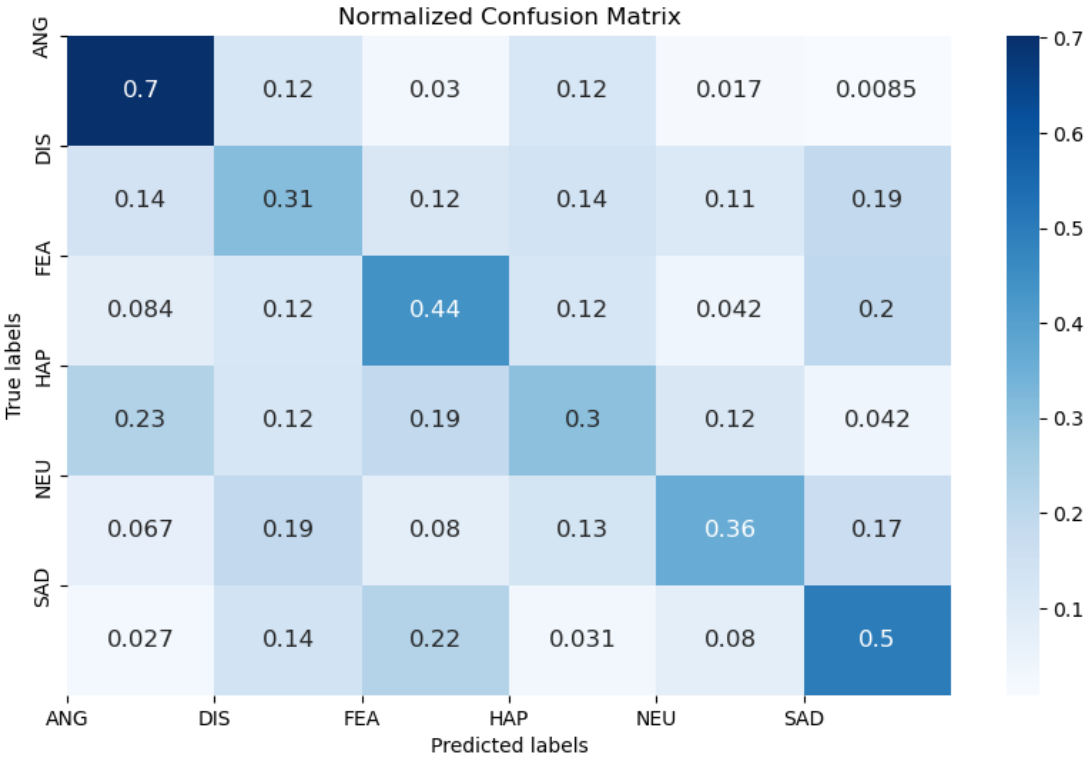


Fig 5.2.4.6 Confusion Matrix for Resnet18-LSTM model – CREMA

C. SAVEE

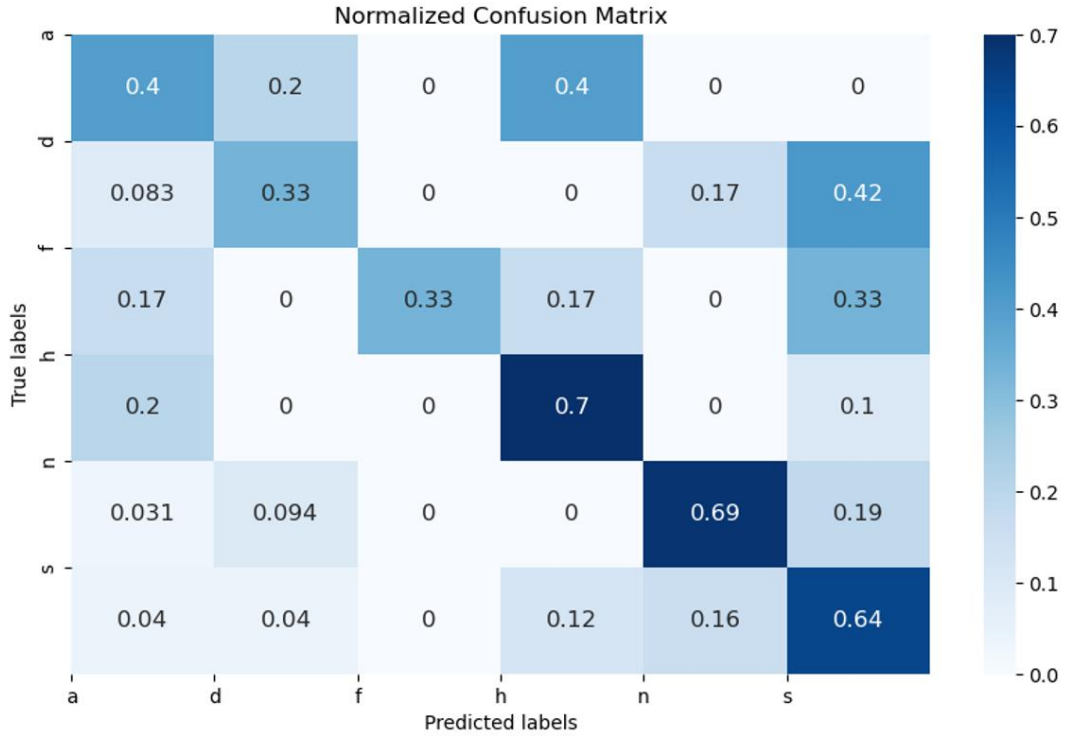


Fig 5.2.4.7 Confusion Matrix for LSTM model - SAVEE

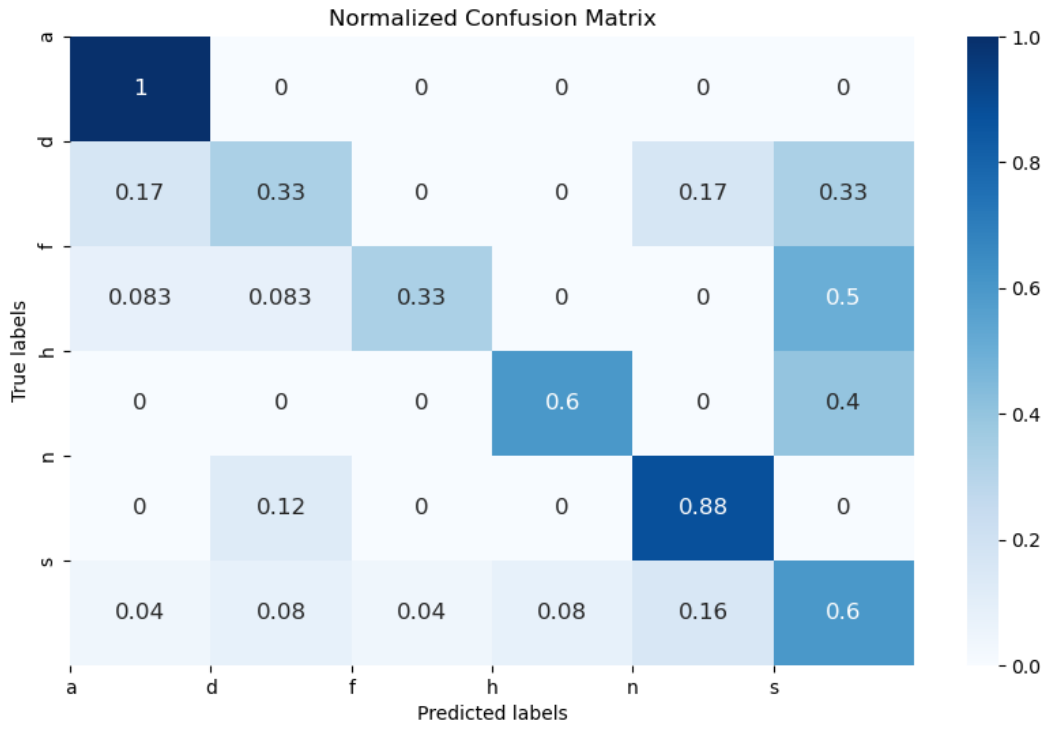


Fig 5.2.4.8 Confusion Matrix for LWCNN-LSTM model – SAVEE

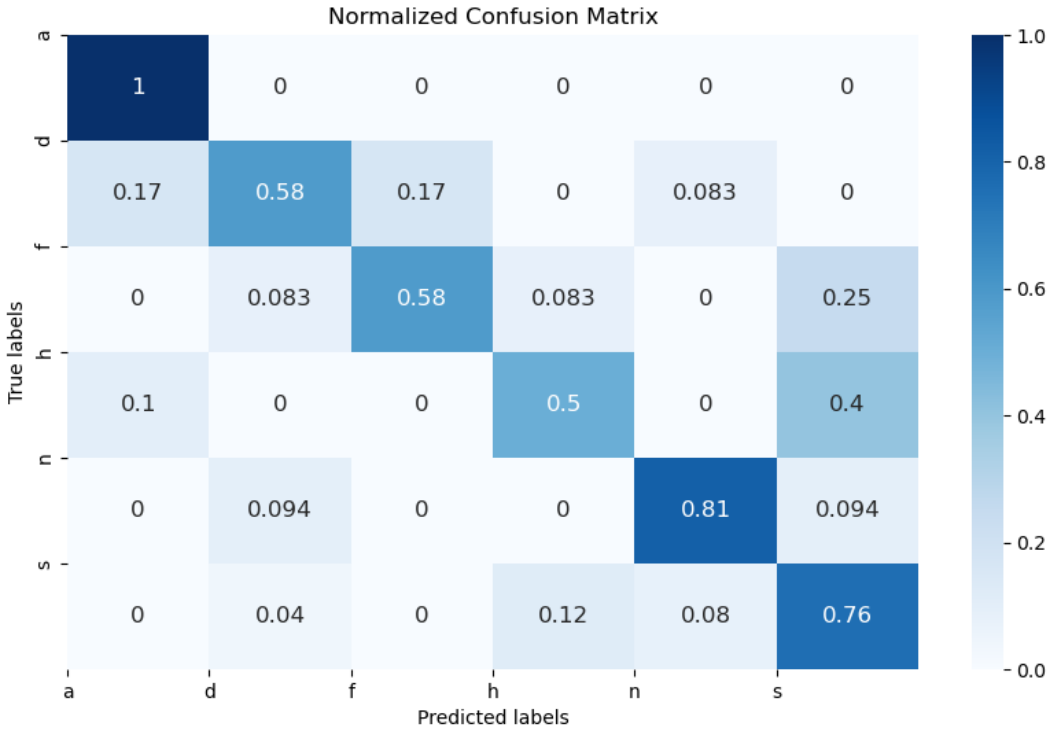


Fig 5.2.4.9 Confusion Matrix for Resnet18-LSTM model – SAVEE

D. RAVDESS

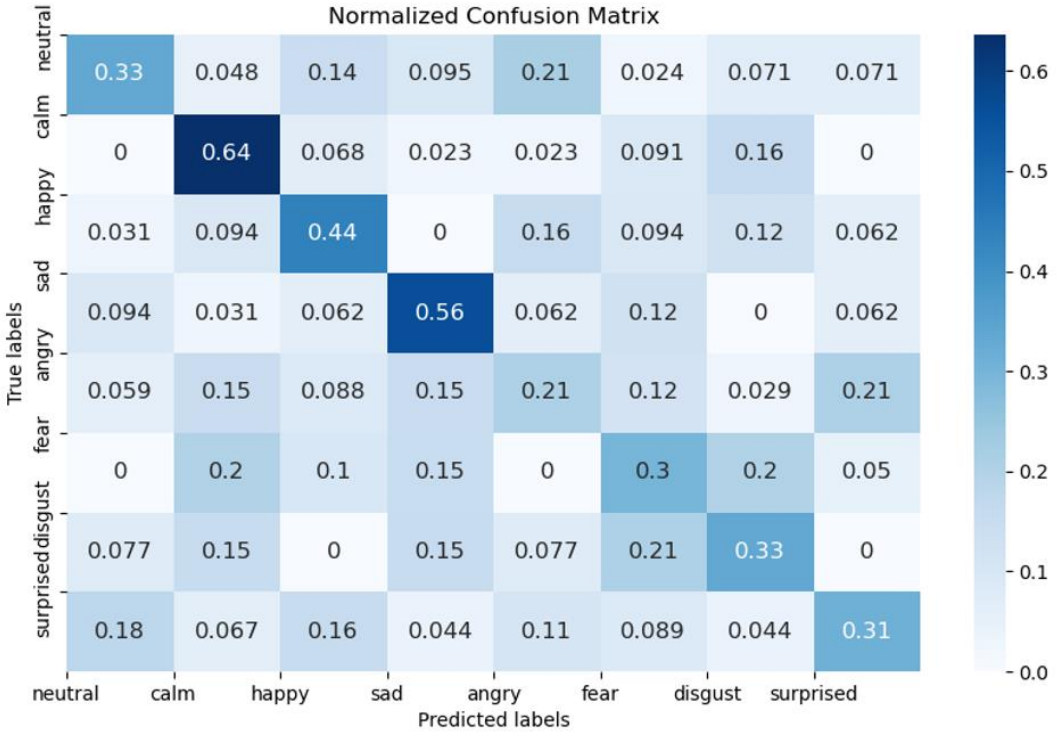


Fig 5.2.4.10 Confusion Matrix for LSTM model - RAVDESS

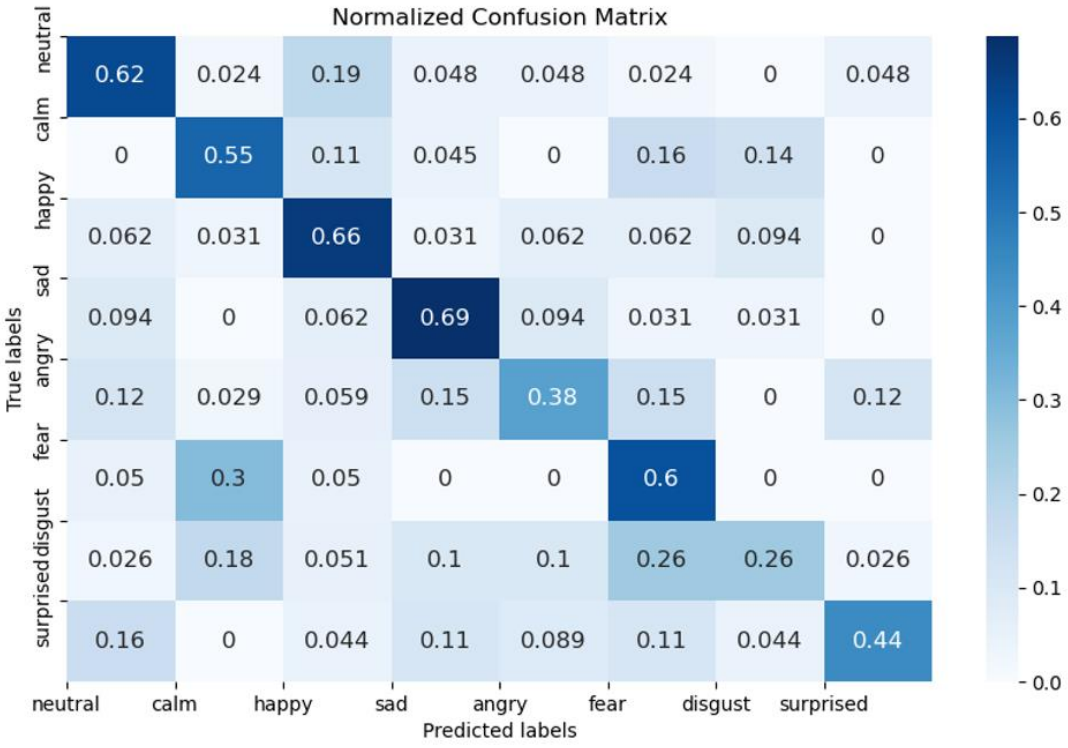


Fig 5.2.4.11 Confusion Matrix for LWCNN-LSTM model – RAVDESS

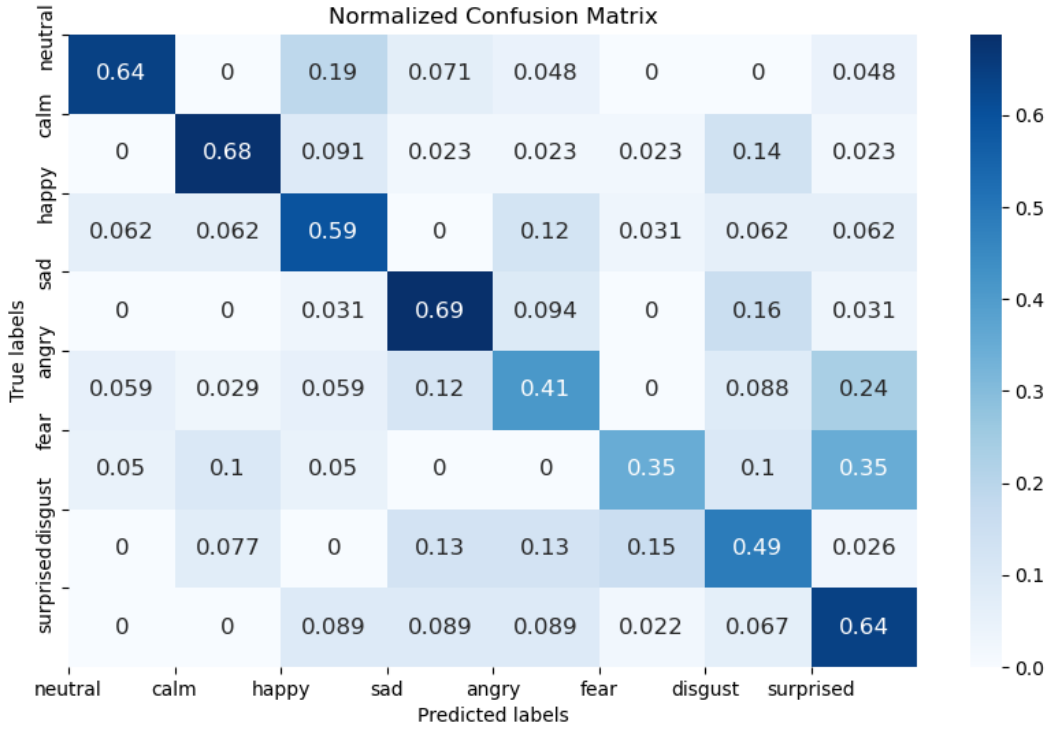


Fig 5.2.4.12 Confusion Matrix for Resnet18-LSTM model – RAVDESS

A notable trend in the validation confusion matrices(Figure 5.2.4.1- 5.2.4.12) is observed across the four datasets: TESS, CREMA, SAVEE and RAVDESS. The validation confusion matrix for the TESS dataset revealed promising results, characterized by a higher level of accuracy. However, when examining the validation confusion matrices for the CREMA, SAVEE and RAVDESS datasets, a contrasting pattern emerged, with values consistently falling within the range of 40 to 80.

This observed trend prompts a deeper exploration into the underlying factors contributing to the discrepancies in classification performance across the different datasets. One plausible explanation is the inherent variability in the characteristics of each dataset. TESS, for instance, emerges as the best-performing dataset, potentially due to attributes such as clearer audio recordings or a more balanced distribution of emotion classes, facilitating more accurate classification compared to CREMA.

In contrast, CREMA consistently demonstrates the worst performance among the datasets, indicating challenges in classification accuracy. Factors such as greater complexity or variability in acoustic features, emotional expressions, or recording conditions could contribute to this discrepancy.

Interestingly, despite the challenges posed by CREMA, the validation confusion matrices for SAVEE and RAVDESS display relatively better performance. This disparity could be attributed to specific characteristics unique to each dataset. For instance, RAVDESS may contain more consistent and controlled emotional expressions due to the use of professional actors, contributing to better classification results. Similarly, SAVEE might benefit from a more focused and controlled experimental setting, resulting in clearer emotional cues and hence better model performance.

Moreover, the distribution of samples within each dataset could also influence model performance. Variations in the number of samples per class, as well as the diversity and representativeness of the emotional expressions captured, may impact the model's ability to generalize effectively across different datasets. Further investigation into these dataset-specific factors is essential for understanding and improving the robustness of speech emotion recognition models across diverse datasets.

Table 5.2.4 Prediction accuracy for different emotions – LSTM Model

EMOTION	ACCURACY (%)			
	TESS	**CREMA**	**SAVEE**	**RAVDESS**
Anger	98.90	71.81	84.61	82.01
Calm	-	-	-	74.02
Disgust	99.45	56.64	80.88	70.80
Fear	100	57.61	87.30	77.02
Happy	98.19	56.83	84.61	73.54
Neutral	100	59.15	77.46	71.69
Surprised	98.90	-	-	73.54
Sad	98.36	65.33	68.75	76.51

Table 5.2.3 presents the accuracy of LSTM model across the four datasets.

Table 5.2.5 Prediction accuracy for different emotions – LWCNN-LSTM model

EMOTION	ACCURACY (%)			
	TESS	CREMA	SAVEE	RAVDESS
Anger	100	77.15	93.93	80.43
Calm	-	-	-	80.43
Disgust	99.82	65.46	80.51	-
Fear	100	72.84	87.32	79.14
Happy	99.64	65.99	91.17	81.76
Neutral	100	71.17	86.11	81.31
Surprised	100	-	-	78.30
Sad	99.46	72.20	72.09	83.61

Table 5.2.4 presents the accuracy of Resnet18-LSTM model across the four datasets.

Table 5.2.6 Prediction accuracy for different emotions – Resnet18-LSTM model

EMOTION	ACCURACY (%)			
	TESS	CREMA	SAVEE	RAVDESS
Anger	100	75.55	95.83	81.86
Calm	-	-	-	89.30
Disgust	100	64.36	87.34	81.06
Fear	100	68.33	90.78	89.30
Happy	99.46	67.11	88.46	84.34
Neutral	100	7306	88.46	90.27
Surprised	100	-	-	82.26
Sad	99.46	72.20	81.17	86.97

Table 5.2.6 presents the accuracy of Resnet18-LSTM model across the four datasets.

From Table 5.2.4 - 5.2.6 we can infer that LSTM model demonstrates strong performance across various emotions, particularly excelling in recognizing anger, fear, and sadness, with accuracies consistently above 98% in most datasets. The LWCNN-LSTM model showcases competitive accuracy rates, notably outperforming LSTM in certain emotions such

as calmness and surprise across specific datasets. These findings suggest that while LSTM architectures exhibit robustness in capturing nuanced emotional expressions, the integration of convolutional layers in the LWCNN-LSTM model offers enhanced sensitivity to certain emotional states, potentially due to its ability to capture spatial features in the audio signals. LWCNN-LSTM architecture likely proved advantageous for this task. CNNs effectively capture spatial features from spectrograms, visual representations of speech that encode frequency information over time. This can be particularly useful for identifying emotions influenced by vocal characteristics. LSTMs, on the other hand, excel at learning long-term dependencies within speech signals, crucial for understanding the emotional context of spoken language. The combination of these strengths in the LWCNN-LSTM architecture likely contributed to the model's success. The ResNet18-LSTM model further extends the capability to recognize various emotions across the datasets. Particularly noteworthy is its exceptional accuracy in identifying anger, disgust, fear, and neutral emotions, where accuracies often reach or exceed 90% across datasets. The model exhibits remarkable consistency in recognizing these emotions, showcasing its robustness and effectiveness in capturing subtle emotional nuances present in speech signals.

The ResNet18-LSTM model demonstrates notable proficiency in recognizing sadness, with accuracies consistently above 80% across datasets. This comprehensive performance underscores the model's ability to capture complex emotional states, thereby offering promising applications in speech emotion recognition tasks across diverse contexts. When juxtaposed with the LSTM and LWCNN-LSTM models, the ResNet18-LSTM model demonstrates its efficacy in achieving high accuracy rates across a wide range of emotions, highlighting its potential as a versatile tool for emotion detection in spoken language.

5.3 POTENTIAL FACTORS INFLUENCING PERFORMANCE

Balanced Emotion Distribution: Emotion recognition accuracy can be significantly influenced by the distribution of emotions within a dataset [15]. Datasets with a balanced representation of all emotion categories (anger, sadness, happiness, fear, surprised, disgust, neutral) can provide a more comprehensive training experience for deep learning models. Conversely, datasets with a skewed distribution towards certain emotions might force the model to prioritize those emotions and perform less well on underrepresented ones. It is crucial to explore the emotion distribution within each dataset (TESS, CREMA, RAVDESS, SAVEE) to determine if this factor contributes to the observed performance variations.

Emotion Elicitation Methods: Speech emotion recognition models are often trained on datasets where emotions are elicited through various methods, such as acting or natural interactions [16]. The TESS dataset might primarily utilize staged emotions delivered by actors. Staged emotions can be more controlled and pronounced compared to spontaneous speech in datasets like CREMA or RAVDESS, which capture emotions expressed during natural conversations. Clear and well-acted emotions might be easier for deep learning models to recognize, potentially leading to the higher performance observed on TESS. Investigating the emotion elicitation methods used in each dataset can provide valuable insights into the performance variations.

Acoustic Characteristics: Speech acoustics can vary depending on the recording environment and speaker characteristics [17]. The TESS dataset might have recordings captured in a controlled studio environment with minimal background noise. In contrast, datasets like CREMA or RAVDESS might include recordings from real-world scenarios with varying noise levels. Background noise and other acoustic variations can hinder the model's ability to extract relevant features for emotion recognition, potentially leading to lower performance. Analyzing the acoustic characteristics (e.g., noise levels, signal-to-noise ratio) of each dataset can be informative.

CHAPTER 6

IMPLEMENTATION

6.1 IMPLEMENTATION

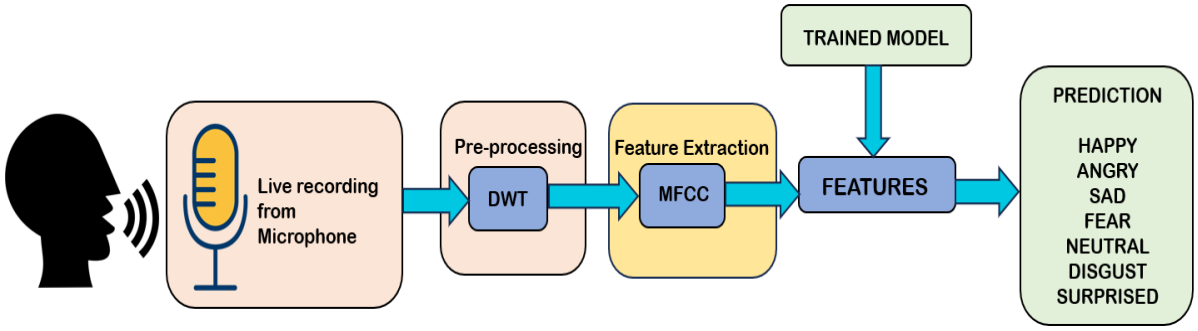


Fig 6.1 Functional Block diagram for Real-Time Prediction

Our real-time implementation seamlessly integrates the LWCNN-LSTM architecture for speech emotion recognition, representing a sophisticated system designed to operate seamlessly on live speech data captured through a microphone interface. The system's operation begins with the preprocessing step, where the incoming speech signals undergo Discrete Wavelet Transform (DWT), allowing for efficient preprocessing. Subsequently, the system leverages Mel-frequency cepstral coefficients (MFCC), a powerful feature extraction technique, tailored to optimize emotion detection across diverse datasets. For our current implementation, we employ the best-performing model parameters derived from the TESS dataset, utilizing DWT level 2 and 40 MFCC coefficients, ensuring superior performance and robustness.

Following the preprocessing and feature extraction stages, the system proceeds to the core of its functionality: the LWCNN-LSTM model. This powerful architecture is adept at processing sequential data, enabling rapid and accurate inference on the emotional content of the input speech. By leveraging both convolutional and recurrent neural network components, the model captures complex temporal and spatial patterns inherent in emotional speech, facilitating nuanced understanding and classification of various emotional states in real-time.

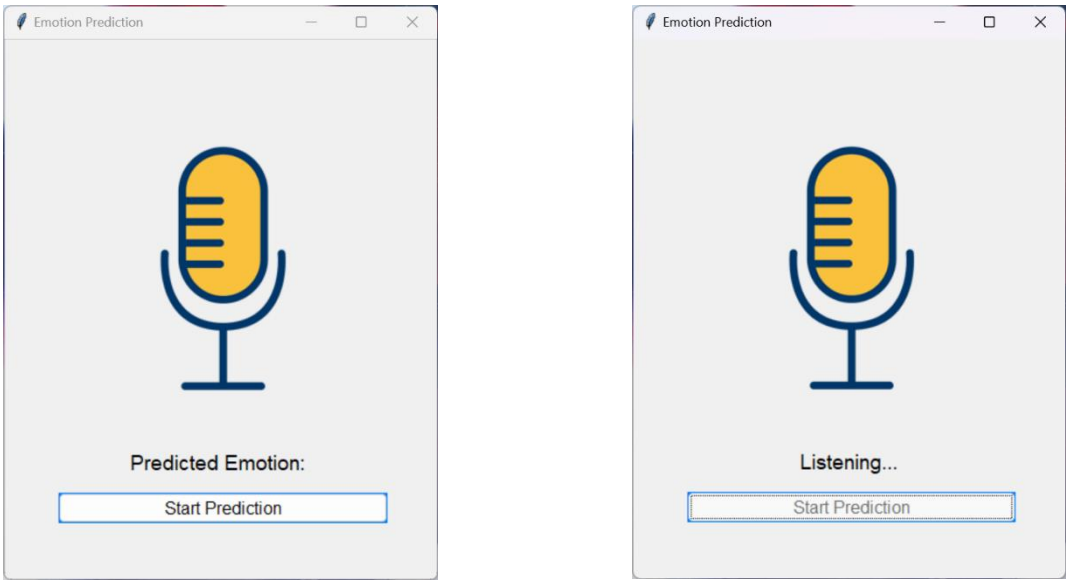


Fig 6.2 Real time prediction using Tkinter – Start Prediction

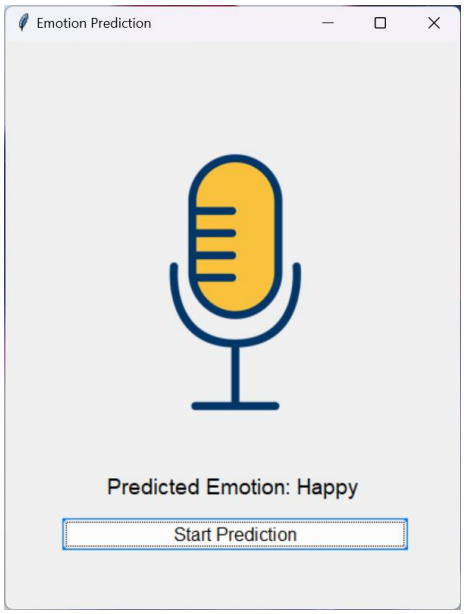


Fig 6.3 Real time prediction using Tkinter – Prediction

Our implementation (Figure 6.2 and Figure 6.3) leverages the tkinter library in Python to develop an intuitive Graphical User Interface (GUI) for real-time prediction tasks. Users interact with the system by inputting audio samples directly through the interface, enabling instant predictions using a LWCNN-LSTM model trained on a diverse dataset. Users can seamlessly interact with the system. After clicking the "Record" button, their voice is captured, and upon clicking "Predict," the LWCNN-LSTM model, trained on a vast and diverse emotional speech dataset, analyzes the audio sample.

The model architecture combines CNN and LSTM networks, adept at capturing both spatial and temporal features present in audio data. To represent the spectral characteristics of

the input signals comprehensively, the model utilizes 40 MFCC coefficients, extracted from the audio samples. Additionally, the signals undergo a Discrete Wavelet Transform (DWT) at level 1, enhancing the model's ability to capture both high and low-frequency components. The predicted emotion is then displayed prominently on the interface, empowering users with real-time insights into their emotional state or the sentiment conveyed in an audio recording. This integration of tkinter-based GUI with sophisticated deep learning techniques empowers users to achieve accurate and efficient real-time predictions.

CHAPTER 7

CONCLUSION

7.1 CONCLUSION

Overall, our findings demonstrate the efficacy of various deep learning architectures for Human Speech Emotion Recognition (HSER), with a focus on the capabilities of LSTM LWCNN-LSTM and Resnet18-LSTM. By conducting extensive experiments on a range of emotional speech datasets, such as CREMA-D, SAVEE, RAVDESS, and TESS, we assessed different model configurations and feature representations, including the Mel-Frequency Cepstral Coefficients (MFCC) and Discrete Wavelet Transform (DWT).

Table 7.1 Performance comparison of the proposed Resnet18_LSTM HSER model with existing models

Author & Year	**Parameter**	**Dataset**	**Model**	**Accuracy/ Precision (%)**
Proposed	**DWT and MFCC**	**TESS**	**Resnet18-LSTM**	**99.82**
Trinh et al [1]	Spectrogram, Mel-frequency	IEMOCAP	GRU	97.47
Wang Et al [6]	Mel-Spectrograms MFCC	IEMOCAP	Dual-Sequence LSTM and CNN	73.3
Xie et al [7]	Mel-frequency	CASIA	Attention Based LSTM	92.82
Kurpukdee et al [8]	Frame level ad utterance level	IEMOCAP	Conv- LSTM	65.36
Rasheed et al [9]	Silent (dB), min-quiet and min-sounding interval	IEMOCAP	CNN Bi-LSTM	87.41
Onah et al [12]	Mel-frequency MFCC	RAVDESS	LSTM and CNN	97
Leelavathi et al [13]	MFCC	RAVDESS	LSTM	78.2
Xiaoyu et al [15]	MFCC	IEMOCAP	CNN-BiLSTM	90.65

The importance of dataset selection in HSER tasks is shown by our findings, which show that models trained on the TESS dataset consistently outperformed those trained on other datasets. Notably, preprocessing using DWT produced significant performance gains, especially for Resnet18-LSTM, LWCNN-LSTM and LSTM designs. In particular, the Resnet18-LSTM model trained with DWT level 4 and N_MFCC 40 demonstrated remarkable accuracy and generalization performance across a range of assessment criteria for the TESS dataset. This model topped all criteria more than 99% and continuously produced outstanding recall, accuracy, and precision values.

In summary, our research offers significant contributions to the development of HSER methodologies by emphasizing the efficiency of deep learning models in obtaining significant characteristics from speech signals. We expect significant improvements in emotion-aware computer applications as we investigate new avenues and continue to hone our models, opening the door to more fluid human-machine interactions.

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